

Zurich ServiceNow AI Platform Capabilities

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Service Graph Connector for GCP

Use the Service Graph Connector for GCP to ingest CMDB data from a Google Cloud Platform (GCP) project using the REST APIs.

Request apps on the Store

Visit the [ServiceNow Store](#) website to view all the available apps and for information about submitting requests to the store. For cumulative release notes information for all released apps, see the [ServiceNow Store version history release notes](#).

Key features

- Get started quickly with a streamlined onboarding process.
- Use only the credentials you need.
- No MID Server required.
- Get end-to-end data coverage.
- No agents or individual scanning needed.
- Performant at scale.

Supported versions

Versions

GCP	ServiceNow
Last tested on July 16, 2025	• Xanadu • Yokohama • Zurich

Use cases

You can use the Service Graph Connector for GCP to get visibility into cloud resource identities, relationships, and state in real-time.

Important information for upgrading Service Graph Connector for GCP

After you've installed the latest version of the Service Graph Connector for GCP, you must synchronize the newly added data sources. Use the Advanced Setup section in the guided setup to synchronize data sources. For more information, see [Configure Service Graph Connector for GCP](#).

Configuring a connection for the connector

You can configure a connection for the connector by using the SGC Central view in the CMDB Workspace. The view enables you to discover and install connectors, and then effectively manage the full life cycle of creating, editing, monitoring, and debugging connections. To configure the connector using SGC Central, see [Configure Service Graph Connector for GCP using SGC Central](#).

Important: Unless there are configuration issues, use the SGC Central view in the CMDB Workspace to configure the connection for the connector, as the guided setup method is being deprecated.

CMDB integrations dashboard

The Integration Commons for CMDB store app provides a dashboard with a central view of the status, processing results, and processing errors of all installed integrations. You can see metrics for all integration runs. You can filter the view to a specific CMDB integration, a specific time duration, or a specific integration run. For more details about monitoring GCP integrations in the CMDB Integrations Dashboard, see [Using the CMDB Integrations Dashboard](#).

Data mapping

Data from the GCP data sources is mapped and transformed into the ServiceNow CMDB Configuration Item (CI) class definitions using the Robust Transform Engine (RTE). Data is inserted into the ServiceNow CMDB using the Identification and Reconciliation Engine (IRE).

When you complete setting up the connection, you can configure the integration to periodically pull data from the GCP application.

The following table lists the data sources, the staging tables, the target tables as the CMDB CI classes, and the resource types imported for a GCP project.

Data mapping for GCP

Data source	Staging table	CMDB CI classes	Resource types
SG-GCP Organization	SG-GCP Organization [sn_gcp_integ_s g_gcp_organiza tion]	Cloud Organizations	Organization
SG-GCP Folder	SG-GCP Folder [sn_gcp_integ_s g_gcp_folder]	Google Organization Folder SG-GCP Extension Attributes	Folder
SG-GCP Project	SG-GCP Project [sn_gcp_integ_s g_gcp_project]	Google Organization Project Cloud Service Account Key Value SG-GCP Extension Attributes	Project
SG-GCP Network	SG-GCP Network [sn_gcp_integ_s g_gcp_network]	Cloud Network	Network

Data source	Staging table	CMDB CI classes	Resource types
		SG-GCP Extension Attributes	
SG-GCP Machine Image	SG-GCP Machine Image [sn_gcp_integ_s g_gcp_machine _image]	Image Key Value SG-GCP Extension Attributes	MachinelImage
SG-GCP Subnet	SG-GCP Subnet [sn_gcp_integ_s g_gcp_subnet]	Cloud Subnet Google Datacenter Key Value SG-GCP Extension Attributes	Subnetwork
SG-GCP Storage Volume	SG-GCP Storage Volume [sn_gcp_integ_s g_gcp_storage_ volume]	Storage Volume Google Datacenter Cloud Disk Type Key Value SG-GCP Extension Attributes	Disk

Data source	Staging table	CMDB CI classes	Resource types
SG-GCP Storage Volume Snapshot	SG-GCP Storage Volume Snapshot [sn_gcp_integ_s g_gcp_storage_ vol_snapshot]	Storage Volume Snapshot Key Value SG-GCP Extension Attributes	Snapshot
SG-GCP Security Group	SG-GCP Security Group [sn_gcp_integ_s g_gcp_security_ group]	Compute Security Group Network ACL Network ACL Rule Key Value SG-GCP Extension Attributes	Firewall
SG-GCP Software Inventory	SG-GCP Software Inventory [sn_gcp_integ_s g_gcp_software_ inventory]	When the Software Asset Management (SAM) application isn't installed: Software Software Instance When the SAM application is installed:	Instance

Data source	Staging table	CMDB CI classes	Resource types
		Software Installation	
SG-GCP VM Instance	SG-GCP VM Instance [sn_gcp_integ_s g_gcp_vm_insta nce]	Virtual Machine Instance Google Datacenter Hardware Type Availability Zone Cloud Mgmt Network Interface Server Storage Mapping VNIC Endpoint Block Endpoint IP Address Key Value SG-GCP Extension Attributes	Instance
SG-GCP Execute Patch Job	SG-GCP Execute Patch Job [sn_gcp_integ_s g_gcp_execute _patch_job]	Not applicable	Not applicable

Data source	Staging table	CMDB CI classes	Resource types
SG-GCP Hardware Type	SG-GCP Hardware Type [sn_gcp_integ_s g_gcp_hardware_type]	Hardware Type Google Datacenter Availability Zone	MachineType
SG-GCP VM Hw Consolidation	SG-GCP VM Hw Consolidation [sn_gcp_integ_s g_gcp_vm_hw_consolidation]	Virtual Machine Instance Server	None
SG-GCP Load Balancer Pool	SG-GCP Load Balancer Pool [sn_gcp_integ_s g_gcp_load_balancer_pool]	Load Balancer Pool Google Datacenter Cloud Service Account SG-GCP Extension Attributes Key Value	TargetPool
SG-GCP Load Balancer Pool Member	SG-GCP Load Balancer Pool Member [sn_gcp_integ_s g_gcp_load_balancer_pool_member]	Load Balancer Pool Member SG-GCP Extension Attributes	TargetPool
SG-GCP Load Balancer Health Service	SG-GCP Load Balancer Health Service[sn_gcp_i	Cloud Load Balancer Health Service	HealthCheck

Data source	Staging table	CMDB CI classes	Resource types
	nteg_sg_gcp_load_balancer_health_service]	SG-GCP Extension Attributes	
SG-GCP Load Balancer	SG-GCP Load Balancer [sn_gcp_integ_sg_gcp_load_balancer]	Cloud Load Balancer Google Datacenter Cloud Service Account SG-GCP Extension Attributes Key Value	BackendService
SG-GCP Load Balancer Service	SG-GCP Load Balancer Service [sn_gcp_integ_sg_gcp_load_balancer_service]	Load Balancer Service SG-GCP Extension Attributes	ForwardingRule
SG-GCP Cloud Database	SG-GCP Cloud Database [sn_gcp_integ_sg_gcp_cloud_database]	Cloud DataBase Google Datacenter SG-GCP Extension Attributes Key Value	Instance

Data source	Staging table	CMDB CI classes	Resource types
SG-GCP Cloud Function	SG-GCP Cloud Function [sn_gcp_integ_s g_gcp_cloud_fu nction]	Cloud Function Google Datacenter SG-GCP Extension Attributes Key Value	Function
SG-GCP Cloud Object Storage	SG-GCP Cloud Object Storage [sn_gcp_integ_s g_gcp_cloud_o bject_storage]	Cloud Object Storage Google Datacenter Key Value SG-GCP Extension Attributes	Buckets
SG-GCP Kubernetes Cluster	SG-GCP Kubernetes Cluster [sn_gcp_integ_s g_gcp_kubern es_cluster]	Kubernetes Cluster Key Value SG-GCP Extension Attributes	Cluster
SG-GCP Kubernetes Node	SG-GCP Kubernetes Node [sn_gcp_integ_s g_gcp_kubern es_node]	Kubernetes Node	Node

Data source	Staging table	CMDB CI classes	Resource types
		SG-GCP Extension Attributes	
SG-GCP Kubernetes Pod	SG-GCP Kubernetes Pod [sn_gcp_integ_s g_gcp_kubern es_pod]	Kubernetes Pod Kubernetes Volume Docker Image Docker Container Key Value SG-GCP Extension Attributes	Pod
SG-GCP Kubernetes Service	SG-GCP Kubernetes Service [sn_gcp_integ_s g_gcp_kubern es_service]	Kubernetes Service Key Value SG-GCP Extension Attributes	Service
SG-GCP Kubernetes Namespace	SG-GCP Kubernetes Namespace [sn_gcp_integ_s g_gcp_kubern es_namespace]	Kubernetes Namespace Cloud Service Account Key Value	Namespace

Data source	Staging table	CMDB CI classes	Resource types
		SG-GCP Extension Attributes	
SG-GCP Kubernetes Deployment	SG-GCP Kubernetes Deployment [sn_gcp_integ_s g_gcp_kubern es_deployment]	Kubernetes Deployment Key Value SG-GCP Extension Attributes	Deployment
SG-GCP Kubernetes Replicaset	SG-GCP Kubernetes Replicaset [sn_gcp_integ_s g_gcp_kubern es_replicaset]	Kubernetes ReplicaSet Key Value SG-GCP Extension Attributes	ReplicaSet
SG-GCP Kubernetes Cluster Roles	SG-GCP Kubernetes Cluster Roles [sn_gcp_integ_s g_gcp_kubern es_cluster_role]	Kubernetes Cluster Role Key Value SG-GCP Extension Attributes	ClusterRole
SG-GCP Kubernetes Cluster Role Binding	SG-GCP Kubernetes Cluster Role Binding [sn_gcp_integ_s g_gcp_kubern	Kubernetes Cluster Role Binding Key Value	ClusterRoleBindi ng

Data source	Staging table	CMDB CI classes	Resource types
	es_cluster_role_binding]	SG-GCP Extension Attributes	
SG-GCP Kubernetes Node Pool	SG-GCP Kubernetes Node Pool [sn_gcp_integ_s g_gcp_kubernet es_node_pool]	Kubernetes Node Pool Key Value SG-GCP Extension Attributes	NodePool
SG-GCP Generic Resource	SG-GCP Generic Resource [sn_gcp_integ_s g_gcp_generic_ resource] Note: The SG-GCP Generic Resource data source only works with a service account that has organizatio n-level access.	Cloud Resource	Resource
SG-GCP Annotation	SG-GCP Annotation [sn_gcp_integ_s g_gcp_annotati on]	Key Value	Annotation

Data source	Staging table	CMDB CI classes	Resource types
SG-GCP Get Patch Job	SG-GCP Get Patch Job [sn_gcp_integ_s g_gcp_get_pat ch_job]	Not applicable	Not applicable
SG-GCP Generic Resource Annotation	SG-GCP Generic Resource Annotation [sn_gcp_integ_s g_gcp_generic_ resource_annot ation]	Key Value	Annotation

- The SG-GCP Annotation data source populates all labels and tags associated with GCP supported resources to the [Key Value](#) table.
- Starting with the Service Graph Connector for GCP 1.10 version, the SG-GCP Generic Resource Annotation data source populates the tags and labels for Generic resources to the [Key Value](#) table.
- The basic information about a GCP resource is stored in the SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes] table.
- After all the data sources are executed, the Server [cmdb_ci_server] class is updated with the class entry, and then based on the Windows or Linux class, the records are created in the Windows Server [cmdb_ci_win_server] and Linux Server [cmdb_ci_linux_server] classes, respectively.
- The SG-GCP Generic Resource data source imports data for generic resources not tracked by other data sources. The connector uses the Service Graph Resource Inclusion Whitelist [sn_cmdb_int_util_service_graph_resource_inclusion_whitelist] table to differentiate between generic and other supported resource types.

The connector first populates all supported resources in the Service Graph Resource Inclusion Whitelist [sn_cmdb_int_util_service_graph_resource_inclusion_whitelist] table. These resources, categorized under their respective supported

resource types, have specific data sources designated for ingestion within the connector. When the SG-GCP Generic Resource data source is executed and retrieves unsupported resource types, they are added to the Service Graph Resource Inclusion Whitelist [sn_cmdb_int_util_service_graph_resource_inclusion_whitelist] table and categorized as generic.

- For any discovered resources deleted later after pulling data, the Service Graph Connector for GCP automatically updates the **Install Status** field of the associated CMDB CI classes to indicate any retired or deleted records. For more information, see [Life cycle management of records in Service Graph Connector for GCP](#) and [Record removal process in Service Graph Connector for GCP](#).
- Starting with the Service Graph Connector for GCP 1.9.0 version, when the sn_itom_pattern.use a single hardware type for cloud data centers system property is set to `true`, hardware type data is added to the Cloud Hardware Type [cmdb_ci_cloud_hardware_type] table, instead of being added to the Hardware Type [cmdb_ci_compute_template] table. For more information, see the [Service Graph Connector For GCP - Migrating to a new hardware type model \[KB1899584\]](#) article in the Now Support Knowledge Base.
- Starting with the Service Graph Connector for GCP 1.9.0 version, when the sn_cmdb_ci_class.use_single_cloud_os_image system property is set to `true`, image type data is added to the Cloud Image [cmdb_ci_cloud_os_image] table, instead of being added to the Image [cmdb_ci_os_template] table. For more information, see the [Service Graph Connector For GCP - Migrating to a new Cloud OS Image model \[KB1983751\]](#) article in the Now Support Knowledge Base.
- Starting with the Service Graph Connector for GCP 1.9.0 version, list API calls return only the information that is required to make batch calls and not the entire resource response.
- Starting with the Service Graph Connector for GCP 1.10 version, parallel data loading is enabled for the SG-GCP Annotation and SG-GCP Generic Resource Annotation data sources.
- Starting with the Service Graph Connector for GCP 1.10 version, you can configure the AppScript Folder IDs connection property to exclude AppScript folder information from imports. Automations are run on AppScript folders, which contain multiple projects that include meta

information. AppScript folders don't contain any resources such as networks, subnets, and VMs.

Specify the AppScript folder IDs as a comma-separated list in the `Value` field of the AppScript Folder IDs connection property to exclude the AppScript folders and projects information from being added to the CMDB during import.

- Starting with the Service Graph Connector for GCP 1.10 version, the SG-GCP Software Inventory scheduled import runs before the SG-GCP VM Instance scheduled import. The SG-GCP Software Inventory scheduled job populates the operating system information for the Server records, which determines the script that needs to be run for deep discovery (Shell script for Linux VMs or PowerShell script for Windows VMs).
- Starting with the Service Graph Connector for GCP 1.10 version, the SG-GCP Get Patch Job scheduled import supports deep discovery by running commands on VMs. For information about deep discovery for the Service Graph Connector for GCP, see the [Enabling deep discovery for Service Graph Connector for GCP](#) article in the Now Support Knowledge Base.

For more information on where data is saved when pulling data from a GCP project, see [CMDB classes targeted in Service Graph Connector for GCP](#).

You can use the IntegrationHub ETL app to view the data maps. See [IntegrationHub ETL](#) for more information.

BYOL and BYOS support

Bringing your own licenses (BYOL) or bringing your own subscriptions (BYOS) is the process of bringing previously purchased on-premises licenses or subscriptions to Google Cloud. The connector supports the BYOL and BYOS of the Image CIs for the Red Hat Enterprise Linux (RHEL) and SUSE Linux Enterprise Server (SLES) services on Google Cloud. The connector populates the Key Value [cmdb_key_value] table with the licensing information. Depending on the platform usage, the key is set to either `Windows_OS_License_Type_automatic` or `Linux_OS_License_Type_automatic`, and the key value is set to either BYOL or BYOS.

Related concepts

- [Service Graph Connectors](#)
- [Handling import failures for large GCP resources](#)
- [Deep discovery for Service Graph Connector for GCP](#)

Configure Service Graph Connector for GCP using SGC Central

Use the playbook available with the SGC Central application to set up the Service Graph Connector for GCP for pulling in GCP project data into the CMDB.

Before you begin

Install Service Graph Connector for GCP version 1.8.0 or later from the ServiceNow Store. For ServiceNow Store installation steps, see [Install a ServiceNow Store application](#).

Role required: The following table shows the roles required for each stage of the playbook.

Role required for each playbook stage

Stage	Role
Prerequisites	admin
Setup	cmdb_inst_admin or admin

About this task

The playbook experience for onboarding connectors is activated with SGC Central in the CMDB Workspace. To configure the SGC Central application, see [Configuring SGC Central](#) and for more information on how to interact with a playbook, see [Interact with Playbook](#).

Procedure

1. Navigate to **Workspaces > CMDB Workspace**.

2. In the CMDB Workspace, select **SGC Central**.
3. On the Overview page, select **Create connection**.
Tip: Alternatively, you can select **Create connection** on the All connections page.
4. On the Create connection window, select the GCP connector type and then select **Create connection**.
5. Complete the initial prerequisites when setting up a connection for the first time using a connector.

Note: This step is required only during the first-time setup. See [Perform initial setup tasks when creating a connection in SGC Central](#).

6. Complete the prerequisites for setting up the GCP environment.
 - a. Set up the GCP environment and create a Java KeyStore (JKS) certificate to encrypt the security certificates obtained from a GCP application.
 - a. In the **Prerequisites** stage of the playbook, select the **Review setup instructions** activity.
 - b. Select **Service Graph connector for GCP - Setup Instructions** link to view the setup instructions included in the [Service Graph connector for GCP - Setup Instructions \[KB1220598\]](#) article in the Now Support Knowledge Base.
 - c. Make a note of the destination keystore password. You need to specify this password while importing the JKS certificate into the Service Graph Connector for GCP application.
 - d. Select the **I have read the setup instructions** check box to confirm that you have set up the GCP environment according to the instructions.
 - e. Select **Continue**.
 - f. After completing the **Review setup instructions** activity, select **Continue**.

- b. Create an X.509 certificate to associate the JKS certificate for the GCP application with the Service Graph Connector for GCP.

Note: You can skip this step if you're reusing an existing certificate by selecting **Skip**.

- a. In the **Prerequisites** stage of the playbook, select the **Create X.509 certificate** activity.
- b. Select **New**.
- c. On the form, fill in the fields.

Create new X.509 Certificate form

Field	Description
Name	Name of the X.509 certificate. For example, SG-GCP-509Certificate-0rg1.
Key store password	Password to access the JKS certificate as noted down in the step 6.a.iii .
Short description	Description of the X.509 certificate.

- d. In the Attachments section, select **Add File** to browse and upload the keystore.p12 file for the JKS certificate you created in step [6.a](#).
 - e. Select the **I have read the setup instructions** check box after you have completed the setup.
 - f. After completing the **Create X.509 certificate** activity, select **Continue**.
7. (Optional) Download the scripts that are required to enable deep discovery for the Service Graph Connector for GCP.
 - a. In the **Prerequisites** stage of the playbook, select the **Download GCP deep discovery scripts** activity.
 - b. Download the deep discovery scripts for Linux and Windows VMs.

For more information about deep discovery for the Service Graph Connector for GCP, see the [Service Graph Connector for GCP - Deep Discovery Setup Instructions \[KB2213095\]](#) article in the Now Support Knowledge Base.

- c. After completing the **Download GCP deep discovery scripts** activity, select **Continue**.
8. Enter connection details and test the API connection for importing GCP data.
 - a. In the **Setup** stage of the playbook, select the **Create and test connection** activity.
 - b. On the form, fill in the fields.

Create and test connection form

Field	Description
Connection Name	Name to identify the GCP connection record.
Service Account Email	ServiceNow service account associated with the GCP application.
Keystore	Name of the X.509 certificate that you created in step 6.b .
Keystore Password	Password to access the JKS certificate as noted down in the step 6.a.iii .
Organization Id	ID of the organization associated with the GCP application.
Discovery Scope	Discovery scope of the GCP application. The available options are:

Field	Description
	Organization Select Organization when the ServiceNow service account has access to data within the GCP organization. Projects Select Projects when the ServiceNow service account has access to data within GCP projects only.

- c. Select **Update and test connection**.
- d. Once the connection test is complete, select **Continue**.
9. (Optional) Configure the deep discovery connection properties.
 - a. In the **Setup** stage of the playbook, select the **Configure connection properties** activity.
If deep discovery is already configured on the connection, the values are pre-populated on the Deep Discovery properties form. For more information about the deep discovery connection properties for GCP, see [Deep discovery connection properties](#).
 - b. Review the pre-populated fields on the Deep Discovery properties form.
 - c. After completing the **Configure connection properties** activity, select **Continue**.
10. Configure the import schedule to import data at regular intervals.
 - a. In the **Setup** stage of the playbook, select the **Configure import schedule** activity.
 - b. Select **Configure import schedule**.
 - c. Expand the Parent scheduled data import within the Import schedules list to select the **SG-GCP Organization** import schedule.

- d. In the Configure import schedule dialog box, select the **Active** check box, and then fill in the run schedule and time details.

For more information, see [Schedule a data import](#).

- e. Select **Save**.
Alternatively, select **Execute Now** to execute the import schedule immediately.
- f. Select **Continue**.

11. In the **Setup** stage of the playbook, select the **Confirm connection creation** activity to verify whether the connection was configured.

What to do next

Select **View all connections** to review the connection details. The configured connection appears in the Installed connections list.

Related concepts

- [Service Graph Connector for GCP](#)
- [Deep discovery for Service Graph Connector for GCP](#)

Configure Service Graph Connector for GCP using the guided setup

Set up scheduled import jobs to pull in data from a GCP project into your CMDB.

Before you begin

Important: Unless there are configuration issues, use the SGC Central view in the CMDB Workspace to configure the connection for the connector, as the guided setup method is being deprecated.

To use this Service Graph Connector, you need a subscription to a Subscription Unit that is based in the IT Operations Management (ITOM) Visibility application or in the ITOM Discovery application. As defined in the section titled "Managed IT Resource Types" in [ServiceNow](#)

[Subscription Unit Overview](#) for your subscription, for managed IT resources that are created or modified in the CMDB by this Service Graph Connector, but that aren't yet managed by [ITOM Visibility](#) or [ITOM Discovery](#), these resources will increase Subscription Unit consumption from that application. Review your current Subscription Unit consumption within ITOM Visibility or ITOM Discovery to ensure available capacity.

Dependencies and requirements:

- The [Integration Commons for CMDB](#) store app, which is automatically installed.
- The CMDB CI class models store app, which is automatically installed. See [CMDB CI Class Models](#).
- The ITOM Discovery License plugin (com.snc.itom.discovery.license). You must activate this plugin.
- ITOM Licensing plugin (com.snc.itom.license). For more information, see [Request Discovery](#).
- The Datastream Action plugin (com.glide.hub.action_type.datastream), which is automatically installed.
- Observability Commons for CMDB (sn_observability), which is only required for event ingestion. This app must be installed prior to installing the connector for Event Management to work. For more information, see [Observability Commons for CMDB](#) on the ServiceNow Store.

Note: If you have an earlier version of the Service Graph Connector for GCP, then don't migrate data from the old connector. You must uninstall the previous version and run the new integration.

Role required: admin

Note: If you create non-admin RunAs users to run scheduled imports, grant the cmdm_inst_admin role to those users.

Procedure

1. Ensure that the application scope is set to the Service Graph Connector for GCP application by using the application picker. For more information, see [Application picker](#).

2. Navigate to **All > Service Graph Connectors > GCP > Setup**.
3. On the Getting started page, select **Get Started**.
4. Configure the connection to send requests to the GCP application.
 - a. In the Configure the Connection and Credentials section of the Service Graph Connector for GCP page, select **Get Started**.
 - b. Set up the GCP environment and create a Java KeyStore (JKS) certificate to encrypt the security certificates obtained from a GCP application.

Make a note of the destination keystore password. You need to specify this password while importing the JKS certificate into the Service Graph Connector for GCP application.

For instructions, see the [Service Graph Connector for GCP - Setup Instructions \[KB1220598\]](#) article in the Now Support Knowledge Base.

Note: After you have set up the GCP environment, return to the guided setup and set the GCP Setup Instructions task in the Configure the Connection and Credentials section to complete by selecting **Mark as Complete**.

- c. Create an X.509 certificate to associate the JKS certificate for the GCP application with the Service Graph Connector for GCP.
 - a. For the Create X.509 certificate task, select **Configure**.
 - b. On the form that opens in a new tab, fill in the fields.

X.509 Certificate form

Field	Description
Name	Name of the X.509 certificate. For example, SG-GCP-509Certificate-0rg1.
Notify on expiration	Users to be notified when the certificate expires. If no users are selected, the logged-in user is added by

Field	Description
	default, along with the last two logged-in users with the administrator role.
Warn in days to expire	Number of days to send a notification before the certificate expires.
Active	Option to activate the certificate.
Type	Certificate container that is automatically set to Java Key Store . Leave the field value set to Java Key Store .
Expires in days	Number of days until the certificate expires.
Key store password	Password to access the JKS certificate as noted down in the previous step .
Short description	Description of the X.509 certificate.

- c. Select the manage attachments icon (.
 - d. Select **Choose file** to browse and upload the keystore.p12 file for the JKS certificate you created in step [4.b](#).
 - e. Close the dialog box.
 - f. Select the **Validate Stores/Certificates** related link.
 - g. On successful validation, select **Update** to return to the guided setup page.
 - h. Set the Create X.509 certificate task to complete by selecting **Mark as Complete**.
- d. Edit the default connection.

- a. For the Edit Default Connection task, select **Configure**.
- b. In the Flow Designer, select **SG-GCP Default Connection**, and then select **Edit**.
- c. On the form that opens in the Edit Connection dialog box, fill in the fields.

Edit Connection form

Field	Description
Service Account Email	Service account created in and associated with the GCP application.
Keystore	Name of the keystore that you created in step 4.b.
Key Password	Destination keystore password associated with the keystore.
Organization Id	ID of the organization associated with the GCP application.
Discovery Scope	Discovery scope of the GCP application. The available options are: organizations Select organizations when the ServiceNow service account has access to data within the GCP organization. projects Select projects when the ServiceNow service account

Field	Description
	has access to data within GCP projects only.

- d. Select **Edit and Get OAuth Token**.
 - e. After the OAuth token flow is completed successfully, close the dialog box and the browser tab, and return to the guided setup page.
 - f. Set the Edit Default Connection task to complete by selecting **Mark as Complete**.
- e. Test the GCP API connection to import data from the GCP application.
- a. For the Test the Connection task, select **Configure** to open the SG-GCP Default Connection page in a new browser tab.
 - b. Test the connection by selecting the **Test Connection** related link.
- Testing the connection may take a few moments. The page is refreshed to show the test results.
- c. When the **Status** field is set to **Success**, close the browser tab and return to the guided setup page.
- If any of the tests have an error code or message or error sign, the connection failed and further troubleshooting is required.
- d. Set the Test the Connection task to complete by selecting **Mark as Complete**.
 - f. Configure the scheduled jobs to import data from the GCP application.
- a. For the Configure the Scheduled Imports task, select **Configure** to open the Scheduled Data Imports list in a new browser tab.
 - b. Select the SG-GCP Organization scheduled job.

- c. On the Scheduled Data Import form, verify the field values for the scheduled job.

For more information, see [Schedule a data import](#).

- d. Select **Active** check box to activate the SG-GCP Organization scheduled job.
- e. In the **Run** field, select **Periodically** to specify how often the job will run.
- f. Select **Update**.
- g. Return to the guided setup page.
- h. Set the Configure the Scheduled Imports task to complete by selecting **Mark as Complete**.

5. (Optional) Add multiple GCP instances.

- a. In the Add Multiple Instances section of the Service Graph Connector for GCP page, select **Get Started**.
- b. Create data sources for the new GCP connection.
 - a. Ensure that you have edit permissions for the Datasource [sys_data_source] table.
 - b. For the Update Data Source Access task, select **Configure**.
 - c. To edit the record, select the **Global** application scope from the application picker.
 - d. In the Application Access related list of the Data Source form that opens in a new tab, select the **Can create**, **Can update**, and **Can delete** check boxes.
 - e. Select **Update**.
 - f. Return to the guided setup page.
 - g. From the application picker, select the **Service Graph Connector for GCP** application scope.
 - h. Set the Update Data Source Access task to complete by selecting **Mark as Complete**.

- c. Create a scheduled import job for the new GCP connection.
 - a. Ensure that you have edit permissions for the Scheduled data import [scheduled_import_set] table.
 - b. For the Update Scheduled Data Import Access task, select **Configure**.
 - c. To edit the record, select the **Global** application scope from the application picker.
 - d. In the Application Access related list of the Scheduled Data Import form that opens in a new tab, select the **Can create**, **Can update**, and **Can delete** check boxes.
 - e. Select **Update**.
 - f. Return to the guided setup page.
 - g. From the application picker, select the **Service Graph Connector for GCP** application scope.
 - h. Set the Update Scheduled Data Import Access task to complete by selecting **Mark as Complete**.
- d. Clear the cache on the Data Source [sys_data_source] and Scheduled Data Imports [scheduled_import_set] tables.
 - a. For the Clear Cache for Data Source and Scheduled Data Imports tables task, select **Configure**.
 - b. In the **Run script** text box of the background script page, enter the following script:

```
GlideTableManager.invalidateTable("sys_data_source");
GlideCacheManager.flushTable("sys_data_source");
GlideTableManager.invalidateTable("scheduled_import_set");
GlideCacheManager.flushTable("scheduled_import_set");
GlideTableManager.invalidateTable("sys_db_object");
GlideCacheManager.flushTable("sys_db_object");
```

- c. Select **Run Script** to run the background script in the **global** scope.

The script may take several minutes to execute.

- d. After the script is executed, select **Close**.

- e. Return to the guided setup page.

- f. From the application picker, select the **Service Graph Connector for GCP** application scope.

- g. Set the Clear Cache for Data Source and Scheduled Data Imports tables task to complete by selecting **Mark as Complete**.

- e. To create an X.509 certificate for the new GCP instance, repeat the step 4.c, and then set the Create X.509 certificate task to complete by selecting **Mark as Complete**.

- f. Add a new connection.

- a. For the Add New Connection task, select **Configure**.

- b. In the Flow Designer, select **Add Connection**.

- c. On the form that opens in the Create Connection dialog box, fill in the fields.

Edit Connection form

Field	Description
Connection Name	Unique name for the new connection.
Service Account Email	ServiceNow service account associated with the GCP application.
Keystore	Name of the keystore that you created in step 5.e.

Field	Description
Key Password	Destination keystore password associated with the keystore.
Organization Id	ID of the organization associated with the GCP application.
Discovery Scope	Discovery scope of the GCP application. The available options are: organizations Select organizations when the ServiceNow service account has access to data within the GCP organization. projects Select projects when the ServiceNow service account has access to data within GCP projects only.

- g. Test the new connection.
 - a. For the Test New Connection task, select **Configure**.
 - b. In Service Graph Connections page that opens in a new browser tab, select the new connection that you want to test from the list of available connections.
 - c. Test the connection by selecting the **Test Connection** related link.

Testing the connection may take a few moments. The page is refreshed to show the test results.

- d. When the **Status** field is set to **Success**, close the browser tab and return to the guided setup page.

If any of the tests have an error code or message or error sign, the connection failed and further troubleshooting is required.

- e. Set the Test New Connection task to complete by selecting **Mark as Complete**.
 - h. To configure the scheduled jobs to import data from the new GCP instance, repeat step 4.f, and then set the Configure the Scheduled Imports task to complete by selecting **Mark as Complete**.
6. Synchronize new data sources to import data from GCP instances.
 - a. In the Advanced Setup section of the Service Graph Connector for GCP page, select **Get Started**.
 - b. Synchronize new data sources for creating all the required credentials to make the new data sources work correctly.
 - a. For the Sync New Data source task, select **Configure**.
 - b. Select **Synchronize Datasources**.
 - c. Return to the guided setup page by closing the Sync New Data source window.
 - d. Set the Sync New Data source task to complete by selecting **Mark as Complete**.
 7. Set up OS Config Patch Jobs to enable deep discovery.
 - a. In the Setup OS Config Patch Jobs for extended discovery section of the Service Graph Connector for GCP page, select **Get Started**.
 - b. For the Download script files task, select **Configure**.
 - c. Save the Shell script and PowerShell script files.

The Shell script contains commands that run on Linux VMs. The PowerShell script contains commands that run on Windows VMs.

- d. In the Shell script and PowerShell script files, specify the name of and the path in the Cloud Storage Bucket where the output of the commands are to be uploaded.
- e. Upload the deep discovery scripts to the Cloud Storage Bucket.
- f. Navigate back to the guided setup page, and set the Download script files task to complete by selecting **Mark as Complete**.
- g. For the Configure GCP Cloud Storage Bucket for running Patch Jobs task, select **Configure**.
- h. To configure the GCP Cloud Storage Bucket to run patch jobs, follow the deep discovery setup instructions in the [Service Graph Connector for GCP - Deep Discovery Setup Instructions \[KB2213095\]](#) article in the Now Support Knowledge Base.
- i. Navigate back to the guided setup page, and set the Configure GCP Cloud Storage Bucket for running Patch Jobs task to complete by selecting **Mark as Complete**.
- j. For the Configure connection properties task, select **Configure**.
- k. On the SG-GCP Configuration Properties form, select the connection for which you want to run deep discovery from the **Connection** list.
In the Properties section of the SG-GCP Configuration Properties form, the following fields are populated with the values for the connection that you selected. If required, update the field values.

Field	Description
Name of the Cloud Storage Bucket where the .sh and .ps1 files are uploaded in Google Cloud Platform	Name of the Cloud Storage Bucket where the deep discovery scripts are saved.
Path to the .sh file uploaded to Cloud Storage Bucket	Path of the deep discovery Shell script file.
Generation number of the .sh file uploaded to Cloud Storage Bucket	Number used to identify the versioning of the cloud storage object.

Field	Description
Path to the .ps1 file uploaded to Cloud Storage Bucket	Path of the deep discovery PowerShell script file.
Generation number of the .ps1 file uploaded to Cloud Storage Bucket	Number used to identify the versioning of the cloud storage object.
Name of the Cloud Storage Bucket provided in .sh and .ps1 files where the outputs of the commands have to be stored	Name of the output Cloud Storage Bucket.
Folder path provided in .sh and .ps1 files where the outputs are uploaded in the Storage Bucket	Path of the output storage folder in the Cloud Storage Bucket.

l. Select **Save**.

m. Navigate back to the guided setup, and set the Configure connection properties task to complete by selecting **Mark as Complete**.

Related tasks

- [Configure Service Graph Connector for GCP using SGC Central](#)

Related concepts

- [Service Graph Connector for GCP](#)
- [Deep discovery for Service Graph Connector for GCP](#)

CMDB classes targeted in Service Graph Connector for GCP

When you complete setting up the connection, you can configure the integration to periodically pull data from a GCP project. The data is saved in tables that extend from the Configuration item [cmdb_ci] table.

Availability Zone [cmdb_ci_availability_zone]

The following attributes in the Availability Zone [cmdb_ci_availability_zone] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id

Relationship created for Availability Zone

Parent class	Relationship type	Child class
Availability Zone [cmdb_ci_availability_zone]	Contains::Contained by	Virtual Machine Instance [cmdb_ci_vm_instance]

Block Endpoint [cmdb_ci_endpoint_block]

The following attributes in the Block Endpoint [cmdb_ci_endpoint_block] table are populated by collected data:

Attribute label	Attribute name
Host	host
Name	name
Install Status	install_status
Operational status	operational_status

Cloud DataBase [cmdb_ci_cloud_database]

The following attributes in the Cloud DataBase [cmdb_ci_cloud_database] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
TCP port(s)	tcp_port
CPU Count	cpu_count
Fully qualified domain name	fqdn
Install Status	install_status
IP Address	ip_address
Memory Size (GB)	memory_size
Node Count	node_count
Operational status	operational_status
State	state
Type	type
Version	version

Relationships created for Cloud DataBase

Parent class	Relationship type	Child class
Cloud DataBase [cmdb_ci_cloud_data base]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_dat acenter]
Cloud DataBase [cmdb_ci_cloud_data base]	Reference	Key Value [cmdb_key_value]

Parent class	Relationship type	Child class
Cloud DataBase [cmdb_ci_cloud_data base]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extensi on_attributes]

Cloud Disk Type [cmdb_ci_disk_type]

The following attributes in the Cloud Disk Type [cmdb_ci_disk_type] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id

Relationships created for Cloud Disk Type

Parent class	Relationship type	Child class
Cloud Disk Type [cmdb_ci_disk_type]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_dat acenter]
Cloud Disk Type [cmdb_ci_disk_type]	Reference	Key Value [cmdb_key_value]

Cloud Function [cmdb_ci_cloud_function]

The following attributes in the Cloud Function [cmdb_ci_cloud_function] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
CodeSha256	codesha256

Attribute label	Attribute name
Function Last Modified	function_last_modified
Install Status	install_status
Language	language
Operational status	operational_status
Version	version

Relationships created for Cloud Function

Parent class	Relationship type	Child class
Cloud Function [cmdb_ci_cloud_function]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]
Cloud Function [cmdb_ci_cloud_function]	Reference	Key Value [cmdb_key_value]
Cloud Function [cmdb_ci_cloud_function]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Cloud Load Balancer [cmdb_ci_cloud_load_balancer]

The following attributes in the Cloud Load Balancer [cmdb_ci_cloud_load_balancer] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Install Status	install_status

Attribute label	Attribute name
Operational status	operational_status
State	state

Relationships created for Cloud Load Balancer

Parent class	Relationship type	Child class
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Reference	Key Value [cmdb_key_value]
Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Cloud Load Balancer Health Service [cmdb_ci_lb_health_service]

The following attributes in the Cloud Load Balancer Health Service [cmdb_ci_lb_health_service] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Healthy threshold	healthy_threshold
Install Status	install_status
Interval in seconds	check_interval_sec
Monitor type protocol	monitor_type

Attribute label	Attribute name
Operational status	operational_status
Port	port
Request path	request_path
Timeout in seconds	timeout_sec
Unhealthy threshold	unhealthy_threshold

Relationships created for Cloud Load Balancer Health Service

Parent class	Relationship type	Child class
Cloud Load Balancer Health Service [cmdb_ci_lb_health_service]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Cloud Load Balancer Health Service [cmdb_ci_lb_health_service]	Contains::Contained by	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]
Cloud Load Balancer Health Service [cmdb_ci_lb_health_service]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Cloud Mgmt Network Interface [cmdb_ci_nic]

The following attributes in the Cloud Mgmt Network Interface [cmdb_ci_nic] table are populated by collected data:

Attribute label	Attribute name
IP Address	ip_address
Name	name

Attribute label	Attribute name
Object ID	object_id
Public IP	public_ip
Install Status	install_status
Operational status	operational_status
Private IP	private_ip
Configuration Item	cmdb_ci

Relationships created for Cloud Mgmt Network Interface

Parent class	Relationship type	Child class
Cloud Mgmt Network Interface [cmdb_ci_nic]	Owns::Owned by	IP Address [cmdb_ci_ip_address]
Cloud Mgmt Network Interface [cmdb_ci_nic]	Reference	Virtual Machine Instance [cmdb_ci_vm_instance]

Cloud Network [cmdb_ci_network]

The following attributes in the Cloud Network [cmdb_ci_network] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Description	short_description
Install Status	install_status

Attribute label	Attribute name
Operational status	operational_status
State	state

Relationships created for Cloud Network

Parent class	Relationship type	Child class
Cloud Network [cmdb_ci_network]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Cloud Network [cmdb_ci_network]	Contains::Contained by	Cloud Subnet [cmdb_ci_cloud_subnet]
Cloud Network [cmdb_ci_network]	Contains::Contained by	Network ACL [cmdb_ci_network_acl]
Cloud Network [cmdb_ci_network]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Cloud Object Storage [cmdb_ci_cloud_object_storage]

The following attributes in the Cloud Object Storage [cmdb_ci_cloud_object_storage] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Cloud Provider	cloud_provider

Attribute label	Attribute name
Install Status	install_status
Operational status	operational_status
Service Name	service_name

Relationships created for Cloud Object Storage

Parent class	Relationship type	Child class
Cloud Object Storage [cmdb_ci_cloud_object_storage]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]
Cloud Object Storage [cmdb_ci_cloud_object_storage]	Reference	Key Value [cmdb_key_value]
Cloud Object Storage [cmdb_ci_cloud_object_storage]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Cloud Organizations [cmdb_ci_cloud_org]

The following attributes in the Cloud Organizations [cmdb_ci_cloud_org] table are populated by collected data:

Attribute label	Attribute name
Object ID	object_id
Install Status	install_status
Name	name
Operational status	operational_status
Time	time

Relationships created for Cloud Organizations

Parent class	Relationship type	Child class
Cloud Organizations [cmdb_ci_cloud_org]	Contains::Contained by	Google Organization Folder [cmdb_ci_gcp_folder]
Cloud Organizations [cmdb_ci_cloud_org]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Cloud Resource [cmdb_ci_cmp_resource]

The following attributes in the Cloud Resource [cmdb_ci_cmp_resource] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Description	short_description
Install status	install_status
Operational status	operational_status
Resource type	resource_type

Relationships created for Cloud Resource

Parent class	Relationship type	Child class
Cloud Resource [cmdb_ci_cmp_resource]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Parent class	Relationship type	Child class
Cloud Resource [cmdb_ci_cmp_resource]	Reference	Key Value [cmdb_key_value]

Cloud Service Account [cmdb_ci_cloud_service_account]

The following attributes in the Cloud Service Account [cmdb_ci_cloud_service_account] table are populated by collected data:

Attribute label	Attribute name
Account Id	account_id
Discovery credentials	discovery_credentials
Name	name
Object ID	object_id
Datacenter Type	datacenter_type
Install status	install_status
Operational status	operational_status
Organization Id	organization_id

Relationships created for Cloud Service Account

Parent class	Relationship type	Child class
Cloud Service Account [cmdb_ci_cloud_service_account]	Reference	Key Value [cmdb_key_value]
Cloud Service Account	Reference	SG-GCP Extension Attributes

Parent class	Relationship type	Child class
[cmdb_ci_cloud_service_account]		[sn_gcp_integ_extension_attributes]

Cloud Subnet [cmdb_ci_cloud_subnet]

The following attributes in the Cloud Subnet [cmdb_ci_cloud_subnet] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Available IP Count	available_ip_count
Broadcast Address	broadcast_address
CIDR	cidr
Gateway	gateway
Install Status	install_status
Operational status	operational_status
State	state
Subnet Mask	subnet_mask

Relationships created for Cloud Subnet

Parent class	Relationship type	Child class
Cloud Subnet [cmdb_ci_cloud_subnet]	Contains::Contained by	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]

Parent class	Relationship type	Child class
Cloud Subnet [cmdb_ci_cloud_subnet]	Reference	Key Value [cmdb_key_value]
Cloud Subnet [cmdb_ci_cloud_subnet]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Compute Security Group [cmdb_ci_compute_security_group]

The following attributes in the Compute Security Group [cmdb_ci_compute_security_group] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Install Status	install_status
Operational status	operational_status
State	state

Relationships created for Compute Security Group

Parent class	Relationship type	Child class
Compute Security Group [cmdb_ci_compute_security_group]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Compute Security Group [cmdb_ci_compute_security_group]	Reference	Key Value [cmdb_key_value]

Parent class	Relationship type	Child class
Compute Security Group [cmdb_ci_compute_security_group]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Docker Container [cmdb_ci_docker_container]

The following attributes in the Docker Container [cmdb_ci_docker_container] table are populated by collected data:

Attribute label	Attribute name
Container id	container_id
Command	command
Container created	container_created_at
Image id	image_id
Install Status	install_status
Name	name
Operational status	operational_status
Status	status

Docker Image [cmdb_ci_docker_image]

The following attributes in the Docker Image [cmdb_ci_docker_image] table are populated by collected data:

Attribute label	Attribute name
Image id	image_id
Name	name

Google Datacenter [cmdb_ci_google_datacenter]

The following attributes in the Google Datacenter [cmdb_ci_google_datacenter] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Region	region
Install Status	install_status

Relationships created for Google Datacenter

Parent class	Relationship type	Child class
Google Datacenter [cmdb_ci_google_datacenter]	Contains::Contained by	Availability Zone [cmdb_ci_availability_zone]
Google Datacenter [cmdb_ci_google_datacenter]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]

Google Organization Folder [cmdb_ci_gcp_folder]

The following attributes in the Google Organization Folder [cmdb_ci_gcp_folder] table are populated by collected data:

Attribute label	Attribute name
Object ID	object_id
Install Status	install_status
Name	name
Operational status	operational_status

Attribute label	Attribute name
Parent	parent_ci
Parent Id	parent_id
Parent Type	parent_type
Time	time

Relationship created for Google Organization Folder

Parent class	Relationship type	Child class
Google Organization Folder [cldb_ci_gcp_folder]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]
Google Organization Folder [cldb_ci_gcp_folder]	Reference	Cloud Organizations [cldb_ci_cloud_org]
Google Organization Folder [cldb_ci_gcp_folder]	Reference	Google Organization Folder [cldb_ci_gcp_folder]

Google Organization Project [cldb_ci_gcp_project]

The following attributes in the Google Organization Project [cldb_ci_gcp_project] table are populated by collected data:

Attribute label	Attribute name
Operational status	operational_status
Discovery credentials	discovery_credentials
Name	name
Object ID	object_id

Attribute label	Attribute name
Install Status	install_status
Parent	parent_ci
Parent Id	parent_id
Parent Type	parent_type
Project Id	project_id
Status	status
Time	time

Relationships created for Google Organization Project

Parent class	Relationship type	Child class
Google Organization Project [cmdb_ci_gcp_project]	Reference	Google Organization Folder [cmdb_ci_gcp_folder]
Google Organization Project [cmdb_ci_gcp_project]	Reference	Cloud Organizations [cmdb_ci_cloud_org]

Hardware Type [cmdb_ci_compute_template]

The following attributes in the Hardware Type [cmdb_ci_compute_template] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Class	className

Attribute label	Attribute name
Description	short_description
Local Storage GB	local_storage_gb
Memory MB	memory_mb
vCPUs	vcpus
Zone	zone

Note: If the `sn_itom_pattern.use` a single hardware type for cloud data centers system property is set to `true`, the `className` is set to Cloud Hardware Type `[cmdb_ci_cloud_hardware_type]`. If the property is set to `false`, the `className` is set to Hardware Type `[cmdb_ci_compute_template]`.

Relationship created for Hardware Type

Parent class	Relationship type	Child class
Hardware Type <code>[cmdb_ci_compute_template]</code>	Hosted on::Hosts	Google Datacenter <code>[cmdb_ci_google_datacenter]</code>

Image `[cmdb_ci_os_template]`

The following attributes in the Image `[cmdb_ci_os_template]` table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Class	className
Fully qualified domain name	fqdn

Attribute label	Attribute name
Install Status	install_status
Operational status	operational_status

Note: If the `sn_cmdb_ci_class.use_single_cloud_os_image` system property is set to `true`, the `className` is set to Cloud Image `[cmdb_ci_cloud_os_image]`. If the property is set to `false`, the `className` is set to Image `[cmdb_ci_os_template]`.

Relationships created for Image

Parent class	Relationship type	Child class
Image [cmdb_ci_os_template]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Image [cmdb_ci_os_template]	Reference	Key Value [cmdb_key_value]
Image [cmdb_ci_os_template]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

IP Address [cmdb_ci_ip_address]

The following attributes in the IP Address `[cmdb_ci_ip_address]` table are populated by collected data:

Attribute label	Attribute name
IP Address	ip_address
Install Status	install_status
IP version	ip_version

Attribute label	Attribute name
Name	name
Operational status	operational_status

Relationship created for IP Address

Parent class	Relationship type	Child class
IP Address [cmdb_ci_ip_address]	Reference	Key Value [cmdb_key_value]

Key Value [cmdb_key_value]

The following attributes in the Key Value [cmdb_key_value] table are populated by collected data:

Attribute label	Attribute name
Key	key
Value	value

The tags and labels associated with the following CI classes are populated in the Key Value [cmdb_key_value] table.

CMDB CI classes	Label populated	Tag populated
Cloud DataBase [cmdb_ci_cloud_data base]	Yes	Yes
Cloud Function [cmdb_ci_cloud_func tion]	Yes	Yes
Cloud Load Balancer [cmdb_ci_cloud_load _balancer]	No	Yes

CMDB CI classes	Label populated	Tag populated
Cloud Load Balancer Health Service [cmdb_ci_lb_health_service]	No	Yes
Cloud Network [cmdb_ci_network]	No	Yes
Cloud Object Storage [cmdb_ci_cloud_object_storage]	Yes	Yes
Cloud Organizations [cmdb_ci_cloud_org]	No	Yes
Cloud Resource [cmdb_ci_cmp_resource]	Yes	Yes
Cloud Service Account [cmdb_ci_cloud_service_account]	Yes	Yes
Cloud Subnet [cmdb_ci_cloud_subnet]	No	Yes
Compute Security Group [cmdb_ci_compute_security_group]	No	Yes
Google Organization Folder [cmdb_ci_gcp_folder]	No	Yes
Google Organization Project	Yes	Yes

CMDB CI classes	Label populated	Tag populated
[cmdb_ci_gcp_project]		
Image [cmdb_ci_os_template]	No	Yes
IP Address [cmdb_ci_ip_address]	Yes	Yes
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Yes	Yes
Kubernetes Cluster Role [cmdb_ci_kubernetes_cluster_role]	Yes	Yes
Kubernetes Cluster Role Binding [cmdb_ci_kubernetes_cluster_role_binding]	Yes	Yes
Kubernetes Deployment [cmdb_ci_kubernetes_deployment]	Yes	Yes
Kubernetes Namespace [cmdb_ci_kubernetes_namespace]	Yes	Yes
Kubernetes Node [cmdb_ci_kubernetes_node]	Yes	Yes
Kubernetes Node Pool [cmdb_ci_kubernetes_node_pool]	Yes	Yes

CMDB CI classes	Label populated	Tag populated
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Yes	Yes
Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]	Yes	Yes
Kubernetes Service [cmdb_ci_kubernetes_service]	Yes	Yes
Load Balancer Pool [cmdb_ci_lb_pool]	No	Yes
Load Balancer Service [cmdb_ci_lb_service]	Yes	Yes
Storage Volume [cmdb_ci_storage_volume]	Yes	Yes
Storage Volume Snapshot [cmdb_ci_storage_volume_snapshot]	Yes	Yes
Virtual Machine Instance [cmdb_ci_vm_instance]	Yes	Yes

Note: Labels are not supported by Google Cloud for the following CI classes:

- Cloud Load Balancer [cmdb_ci_cloud_load_balancer]
- Cloud Load Balancer Health Service [cmdb_ci_lb_health_service]
- Cloud Network [cmdb_ci_network]
- Cloud Organizations [cmdb_ci_cloud_org]
- Cloud Subnet [cmdb_ci_cloud_subnet]
- Compute Security Group [cmdb_ci_compute_security_group]
- Google Organization Folder [cmdb_ci_gcp_folder]
- Image [cmdb_ci_os_template]
- Load Balancer Pool [cmdb_ci_lb_pool]

Kubernetes Cluster [cmdb_ci_kubernetes_cluster]

The following attributes in the Kubernetes Cluster [cmdb_ci_kubernetes_cluster] table are populated by collected data:

Attribute label	Attribute name
Cluster Resource ID	cluster_resource_id
IP Address	ip_address
Kubernetes UID	k8s_uid
Name	name
Port	port
Cluster Version	cluster_version
Description	short_description

Attribute label	Attribute name
Install Status	install_status
Operational status	operational_status
SelfLink	self_link
Subnet Mask	subnet_mask

Relationships created for Kubernetes Cluster

Parent class	Relationship type	Child class
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Cluster of::Cluster	Kubernetes Node [cmdb_ci_kubernetes_node]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Contains::Contained by	Kubernetes Pod [cmdb_ci_kubernetes_pod]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Contains::Contained by	Kubernetes Cluster Role [cmdb_ci_kubernetes_cluster_role]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Contains::Contained by	Kubernetes Cluster Role Binding [cmdb_ci_kubernetes_cluster_role_binding]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Contains::Contained by	Kubernetes Namespace [cmdb_ci_kubernetes_namespace]

Parent class	Relationship type	Child class
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Contains::Contained by	Kubernetes Service [cmdb_ci_kubernetes_service]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Reference	Key Value [cmdb_key_value]
Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Kubernetes Cluster Role [cmdb_ci_kubernetes_cluster_role]

The following attributes in the Kubernetes Cluster Role [cmdb_ci_kubernetes_cluster_role] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Comments	comments
Description	short_description
Install Status	install_status
Operational status	operational_status
Kubernetes Cluster	cluster
Name	name

Relationships created for Kubernetes Cluster Role

Parent class	Relationship type	Child class
Kubernetes Cluster Role [cmdb_ci_kubernetes_cluster_role]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]
Kubernetes Cluster Role [cmdb_ci_kubernetes_cluster_role]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Kubernetes Cluster Role [cmdb_ci_kubernetes_cluster_role]	Reference	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]
Kubernetes Cluster Role [cmdb_ci_kubernetes_cluster_role]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]
Kubernetes Cluster Role [cmdb_ci_kubernetes_cluster_role]	Reference	Key Value [cmdb_key_value]

Kubernetes Cluster Role Binding [cmdb_ci_kubernetes_cluster_role_binding]

The following attributes in the Kubernetes Cluster Role Binding [cmdb_ci_kubernetes_cluster_role_binding] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Comments	comments

Attribute label	Attribute name
Description	short_description
Install Status	install_status
Name	name
Kubernetes Cluster	cluster
Operational status	operational_status

Relationships created for Kubernetes Cluster Role Binding

Parent class	Relationship type	Child class
Kubernetes Cluster Role Binding [cmdb_ci_kubernetes_cluster_role_binding]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]
Kubernetes Cluster Role Binding [cmdb_ci_kubernetes_cluster_role_binding]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Kubernetes Cluster Role Binding [cmdb_ci_kubernetes_cluster_role_binding]	Reference	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]
Kubernetes Cluster Role Binding [cmdb_ci_kubernetes_cluster_role_binding]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]
Kubernetes Cluster Role Binding [cmdb_ci_kubernetes_cluster_role_binding]	Reference	Key Value [cmdb_key_value]

Kubernetes Deployment [cmdb_ci_kubernetes_deployment]

The following attributes in the Kubernetes Deployment [cmdb_ci_kubernetes_deployment] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Namespace	namespace
Available Replicas	available_replicas
Description	short_description
Desired Replicas	desired_replicas
Install Status	install_status
Kubernetes Cluster	cluster
Operational status	operational_status
Total Replicas	total_replicas
Unavailable Replicas	unavailable_replicas
Updated Replicas	updated_replicas

Relationships created for Kubernetes Deployment

Parent class	Relationship type	Child class
Kubernetes Deployment [cmdb_ci_kubernetes_deployment]	Hosted on::Hosts	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]
Kubernetes Deployment	Hosted on::Hosts	Cloud Service Account

Parent class	Relationship type	Child class
[cmdb_ci_kubernetes_deployment]		[cmdb_ci_cloud_service_account]
Kubernetes Deployment [cmdb_ci_kubernetes_deployment]	Reference	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]
Kubernetes Deployment [cmdb_ci_kubernetes_deployment]	Reference	Key Value [cmdb_key_value]
Kubernetes Deployment [cmdb_ci_kubernetes_deployment]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Kubernetes Namespace [cmdb_ci_kubernetes_namespace]

The following attributes in the Kubernetes Namespace [cmdb_ci_kubernetes_namespace] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Description	short_description
Install Status	install_status
Namespace	namespace
Operational status	operational_status

Relationships created for Kubernetes Namespace

Parent class	Relationship type	Child class
Kubernetes Namespace [cmdb_ci_kubernetes_namespace]	Contains::Contained by	Kubernetes Service [cmdb_ci_kubernetes_service]
Kubernetes Namespace [cmdb_ci_kubernetes_namespace]	Contains::Contained by	Kubernetes Deployment [cmdb_ci_kubernetes_deployment]
Kubernetes Namespace [cmdb_ci_kubernetes_namespace]	Contains::Contained by	Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]
Kubernetes Namespace [cmdb_ci_kubernetes_namespace]	Contains::Contained by	Kubernetes Pod [cmdb_ci_kubernetes_pod]
Kubernetes Namespace [cmdb_ci_kubernetes_namespace]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Kubernetes Namespace [cmdb_ci_kubernetes_namespace]	Reference	Key Value [cmdb_key_value]
Kubernetes Namespace [cmdb_ci_kubernetes_namespace]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Kubernetes Node [cmdb_ci_kubernetes_node]

The following attributes in the Kubernetes Node [cmdb_ci_kubernetes_node] table are populated by collected data:

Attribute label	Attribute name
IP Address	ip_address
Kubernetes UID	k8s_uid
Name	name
Description	short_description
Install Status	install_status
Namespace	namespace
Operational status	operational_status
Kubernetes Cluster	cluster

Relationships created for Kubernetes Node

Parent class	Relationship type	Child class
Kubernetes Node [cmdb_ci_kubernetes_node]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]
Kubernetes Node [cmdb_ci_kubernetes_node]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Kubernetes Node [cmdb_ci_kubernetes_node]	Reference	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]
Kubernetes Node [cmdb_ci_kubernetes_node]	Reference	SG-GCP Extension Attributes [sn_gcp_integration_attributes]

Parent class	Relationship type	Child class
Kubernetes Node [cmdb_ci_kubernetes_node]	Reference	Key Value [cmdb_key_value]

Kubernetes Node Pool [cmdb_ci_kubernetes_node_pool]

The following attributes in the Kubernetes Node Pool [cmdb_ci_kubernetes_node_pool] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Disk Type	disk_type
Image Type	image_type
Install Status	install_status
Machine Type	machine_type
Operational status	operational_status
Pod CIDR Range	podIpv4CidrSize
Resource version	resource_version
Kubernetes Cluster	cluster

Relationships created for Kubernetes Node Pool

Parent class	Relationship type	Child class
Kubernetes Node Pool [cmdb_ci_kubernetes_node_pool]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]

Parent class	Relationship type	Child class
Kubernetes Node Pool [cmdb_ci_kubernetes_node_pool]	Uses::Used by	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]
Kubernetes Node Pool [cmdb_ci_kubernetes_node_pool]	Reference	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]
Kubernetes Node Pool [cmdb_ci_kubernetes_node_pool]	Reference	Key Value [cmdb_key_value]
Kubernetes Node Pool [cmdb_ci_kubernetes_node_pool]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Kubernetes Pod [cmdb_ci_kubernetes_pod]

The following attributes in the Kubernetes Pod [cmdb_ci_kubernetes_pod] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Namespace	namespace
Description	short_description
Host IP	host_ip
Install Status	install_status
IP Address	ip_address
Operational status	operational_status

Attribute label	Attribute name
Resource version	resource_version

Relationships created for Kubernetes Pod

Parent class	Relationship type	Child class
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Contains::Contained by	Docker Image [cmdb_ci_docker_image]
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Contains::Contained by	Kubernetes Volume [cmdb_ci_kubernetes_volume]
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Contains::Contained by	Docker Container [cmdb_ci_docker_container]
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Reference	Key Value [cmdb_key_value]
Kubernetes Pod [cmdb_ci_kubernetes_pod]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]

The following attributes in the Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Namespace	namespace
Description	short_description
Desired Replicas	desired_replicas
Install Status	install_status
Operational status	operational_status
SelfLink	self_link
Total Replicas	total_replicas
Kubernetes Cluster	cluster

Relationships created for Kubernetes ReplicaSet

Parent class	Relationship type	Child class
Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_service_account]
Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]	Hosted on::Hosts	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]
Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]	Reference	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]

Parent class	Relationship type	Child class
Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]	Reference	Key Value [cmdb_key_value]
Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extensi on_attributes]

Kubernetes Service [cmdb_ci_kubernetes_service]

The following attributes in the Kubernetes Service [cmdb_ci_kubernetes_service] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Name	name
Namespace	namespace
Description	short_description
Install Status	install_status
IP Address	ip_address
Operational status	operational_status

Relationships created for Kubernetes Service

Parent class	Relationship type	Child class
Kubernetes Service [cmdb_ci_kubernetes_service]	Hosted on::Hosts	Cloud Service Account [cmdb_ci_cloud_servi ce_account]

Parent class	Relationship type	Child class
Kubernetes Service [cmdb_ci_kubernetes_service]	Reference	Key Value [cmdb_key_value]
Kubernetes Service [cmdb_ci_kubernetes_service]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Kubernetes Volume [cmdb_ci_kubernetes_volume]

The following attributes in the Kubernetes Volume [cmdb_ci_kubernetes_volume] table are populated by collected data:

Attribute label	Attribute name
Kubernetes UID	k8s_uid
Mount Path	mount_path
Name	name
Volume ID	volume_id
Namespace	namespace

Load Balancer Pool [cmdb_ci_lb_pool]

The following attributes in the Load Balancer Pool [cmdb_ci_lb_pool] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Install Status	install_status

Attribute label	Attribute name
Operational status	operational_status

Relationships created for Load Balancer Pool

Parent class	Relationship type	Child class
Load Balancer Pool [cmdb_ci_lb_pool]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]
Load Balancer Pool [cmdb_ci_lb_pool]	Owns::Owned by	Load Balancer Pool Member [cmdb_ci_lb_pool_member]
Load Balancer Pool [cmdb_ci_lb_pool]	Reference	Key Value [cmdb_key_value]
Load Balancer Pool [cmdb_ci_lb_pool]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Load Balancer Pool Member [cmdb_ci_lb_pool_member]

The following attributes in the Load Balancer Pool Member [cmdb_ci_lb_pool_member] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Install status	install_status
Operational status	operational_status

Relationship created for Load Balancer Pool Member

Parent class	Relationship type	Child class
Load Balancer Pool Member [cmdb_ci_lb_pool_member]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Load Balancer Service [cmdb_ci_lb_service]

The following attributes in the Load Balancer Service [cmdb_ci_lb_service] table are populated by collected data:

Attribute label	Attribute name
IP Address	ip_address
Name	name
Object ID	object_id
Port	port
Install Status	install_status
Listener Protocol	listener_protocol
Operational status	operational_status
Service Type	service_type

Relationships created for Load Balancer Service

Parent class	Relationship type	Child class
Load Balancer Service [cmdb_ci_lb_service]	Contains::Contained by	Load Balancer Pool [cmdb_ci_lb_pool]
Load Balancer Service [cmdb_ci_lb_service]	Hosted on::Hosts	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]

Parent class	Relationship type	Child class
Load Balancer Service [cmdb_ci_lb_service]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extensi on_attributes]
Load Balancer Service [cmdb_ci_lb_service]	Reference	Key Value [cmdb_key_value]

Network ACL [cmdb_ci_network_acl]

The following attributes in the Network ACL [cmdb_ci_network_acl] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Description	short_description
Install Status	install_status
Operational status	operational_status

Relationships created for Network ACL

Parent class	Relationship type	Child class
Network ACL [cmdb_ci_network_acl]	Contains::Contained by	Network ACL Rule [cmdb_ci_network_acl _rule]
Network ACL [cmdb_ci_network_acl]	Reference	Key Value [cmdb_key_value]

Network ACL Rule [cmdb_ci_network_acl_rule]

The following attributes in the Network ACL Rule [cmdb_ci_network_acl_rule] table are populated by collected data:

Attribute label	Attribute name
Name	name
Allow Deny	allow_deny
Allowed\Denied Traffic	allowed_denied_traffic
Destination Ranges	destination_ranges
Install Status	install_status
Operational status	operational_status
Outbound	is_outbound
Source Ranges	source_ranges
Target Tags	target_tags

Relationship created for Network ACL Rule

Parent class	Relationship type	Child class
Network ACL Rule [cmdb_ci_network_acl_rule]	Reference	Key Value [cmdb_key_value]

Server [cmdb_ci_server]

The following attributes in the Server [cmdb_ci_server] table are populated by collected data:

Attribute label	Attribute name
Name	name

Attribute label	Attribute name
Object ID	object_id
Class	sys_class_name
CPU count	cpu_count
Disk space (GB)	disk_space
Host name	host_name
Install Status	install_status
IP Address	ip_address
Is Virtual	virtual
Operating System	os
Operational status	operational_status
OS Version	os_version
RAM (MB)	ram

Note: If the GCP Systems Manager (SSM) service isn't enabled, the connector populates the server records in the Server [cmdb_ci_server] class. If the GCP SSM service is enabled, then based on the platform type obtained through the SSM service, the server records are populated in either the Linux Server [cmdb_ci_linux_server] class or the Windows Server [cmdb_ci_win_server] class. The Server [cmdb_ci_server] class is the parent class of the Linux Server [cmdb_ci_linux_server] and the Windows Server [cmdb_ci_win_server] classes.

Relationships created for Server

Parent class	Relationship type	Child class
Server [cmdb_ci_server]	Virtualized by::Virtualizes	Virtual Machine Instance

Parent class	Relationship type	Child class
		[cmdb_ci_vm_instance]
Server [cmdb_ci_server]	Owns::Owned by	IP Address [cmdb_ci_ip_address]
Server [cmdb_ci_server]	Reference	Software Installation [cmdb_sam_sw_install]
Server [cmdb_ci_server]	Reference	Key Value [cmdb_key_value]
Server [cmdb_ci_server]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

The following attributes in the SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes] table are populated by collected data:

Attribute label	Attribute name
Resource ID	resource_id
Folder ID	folder_id
Organization ID	organization_id
Parent	parent
Project	project
Project ID	project_id
Self Link	self_link
Service Graph Connection Record	sg_connection_record

Attribute label	Attribute name
Region	region
Resource Type	resource_type
Organization-Credential Record	org_credential_record

Software [cmdb_ci_spkg]

The following attributes in the Software [cmdb_ci_spkg] table are populated by collected data when the Software Asset Management (SAM) application isn't installed:

Attribute label	Attribute name
Key	key
Discovery source	discovery_source
Name	name
Version	version
Manufacturer	manufacturer

Relationship created for Software

Parent class	Relationship type	Child class
Software [cmdb_ci_spkg]	Reference	Software Instance [cmdb_software_instance]

Software Installation [cmdb_sam_sw_install]

The following attributes in the Software Installation [cmdb_sam_sw_install] table are populated by collected data when the SAM application is installed:

Attribute label	Attribute name
Display name	display_name
Publisher	publisher
Version	version
Discovery source	discovery_source
Last scanned	last_scanned
Installed on	installed_on

Relationship created for Software Installation

Parent class	Relationship type	Child class
Software Installation [cmdb_sam_sw_install]	Reference	Server [cmdb_ci_server]

Software Instance [cmdb_software_instance]

The following attributes in the Software Instance [cmdb_software_instance] table are populated by collected data when the SAM application isn't installed:

Attribute label	Attribute name
Name	name
Installed on	installed_on
Install date	install_date

Relationship created for Software Instance

Parent class	Relationship type	Child class
Software Instance [cmdb_software_instance]	Reference	Server [cmdb_ci_server]

Storage Mapping [cmdb_ci_storage_mapping]

The following attributes in the Storage Mapping [cmdb_ci_storage_mapping] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Install Status	install_status
Mapping Type	mapping_type
Operational status	operational_status

Storage Volume [cmdb_ci_storage_volume]

The following attributes in the Storage Volume [cmdb_ci_storage_volume] table are populated by collected data:

Attribute label	Attribute name
Object ID	object_id
Volume ID	volume_id
Install Status	install_status
Name	name
Operational status	operational_status

Attribute label	Attribute name
Size bytes	size_bytes
State	state
Storage type	storage_type

Relationships created for Storage Volume

Parent class	Relationship type	Child class
Storage Volume [cmdb_ci_storage_volume]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]
Storage Volume [cmdb_ci_storage_volume]	Reference	Key Value [cmdb_key_value]
Storage Volume [cmdb_ci_storage_volume]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

Storage Volume Snapshot [cmdb_ci_storage_vol_snapshot]

The following attributes in the Storage Volume Snapshot [cmdb_ci_storage_vol_snapshot] table are populated by collected data:

Attribute label	Attribute name
Name	name
Object ID	object_id
Capacity	capacity
Install Status	install_status
Operational status	operational_status

Attribute label	Attribute name
Parent ID	parent_id
Size (GB)	size
State	state
Volume Name	volume_name

Relationships created for Storage Volume Snapshot

Parent class	Relationship type	Child class
Storage Volume Snapshot [cmdb_ci_storage_vol_snapshot]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_dat acenter]
Storage Volume Snapshot [cmdb_ci_storage_vol_snapshot]	Provisioned From::Provisioned	Storage Volume [cmdb_ci_storage_vol ume]
Storage Volume Snapshot [cmdb_ci_storage_vol_snapshot]	Reference	Key Value [cmdb_key_value]
Storage Volume Snapshot [cmdb_ci_storage_vol_snapshot]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extensi on_attributes]

Virtual Machine Instance [cmdb_ci_vm_instance]

The following attributes in the Virtual Machine Instance [cmdb_ci_vm_instance] table are populated by collected data:

Attribute label	Attribute name
Name	name

Attribute label	Attribute name
Object ID	object_id
CPUs	cpus
Disks	disks
Disks size (GB)	disks_size
Install Status	install_status
IP Address	ip_address
Memory (MB)	memory
Network adapters	nics
Operational status	operational_status
State	state
Termination Protection	termination_protection
VM Instance ID	vm_inst_id

Relationships created for Virtual Machine Instance

Parent class	Relationship type	Child class
Virtual Machine Instance [cmdb_ci_vm_instance]	Provisioned From::Provisioned	Hardware Type [cmdb_ci_compute_template]
Virtual Machine Instance [cmdb_ci_vm_instance]	Contains::Contained by	Cloud Subnet [cmdb_ci_cloud_subnet]

Parent class	Relationship type	Child class
Virtual Machine Instance [cmdb_ci_vm_instance]	Hosted on::Hosts	Google Datacenter [cmdb_ci_google_datacenter]
Virtual Machine Instance [cmdb_ci_vm_instance]	Contains::Contained by	Cloud Mgmt Network Interface [cmdb_ci_nic]
Virtual Machine Instance [cmdb_ci_vm_instance]	Contains::Contained by	Storage Mapping [cmdb_ci_storage_mapping]
Virtual Machine Instance [cmdb_ci_vm_instance]	Provisioned From::Provisioned	Image [cmdb_ci_os_template]
Virtual Machine Instance [cmdb_ci_vm_instance]	Use End Point To::Use End Point From	Block Endpoint [cmdb_ci_endpoint_block]
Virtual Machine Instance [cmdb_ci_vm_instance]	Hosted on::Hosts	Google Organization Project [cmdb_ci_gcp_project]
Virtual Machine Instance [cmdb_ci_vm_instance]	Reference	Key Value [cmdb_key_value]
Virtual Machine Instance [cmdb_ci_vm_instance]	Reference	SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes]

VNIC Endpoint [cmdb_ci_endpoint_vnic]

The following attributes in the VNIC Endpoint [cmdb_ci_endpoint_vnic] table are populated by collected data:

Attribute label	Attribute name
Host	host
Name	name
Object ID	object_id
Install Status	install_status
IP Address	ip_address
Operational status	operational_status

Relationship created for VNIC Endpoint

Parent class	Relationship type	Child class
VNIC Endpoint [cmdb_ci_endpoint_vnic]	Implement End Point To::Implement End Point From	Cloud Mgmt Network Interface [cmdb_ci_nic]

GCP discovery solutions comparison

ITOM Visibility applications discover a variety of Google Cloud Platform (GCP) resources and populate the relevant configuration item (CI) classes in the Configuration Management Database (CMDB) with their attributes.

ITOM Visibility solutions

ITOM Visibility can offer you comprehensive visibility into your organization's cloud infrastructure. Each application offers unique strengths and capabilities, and by comparing them you can gain holistic insights on their capabilities and benefits.

ITOM Visibility solutions for discovering GCP

ITOM Visibility solution	Overview	Configuration
Discovery and Service Mapping Patterns and Visibility Content	Discovery patterns used by ITOM Visibility	- You must update your store apps to the latest release version. - You must provide ITOM Visibility with access permissions. For more information, see Set up Google Cloud Platform service accounts
Service Graph Connector for GCP	Service Graph Connector for GCP	Configure Service Graph Connector for GCP using the guided setup

Note:

The CI class, in most cases, is applicable to both solutions. If it's not applicable to one of the solutions, the solution is marked "N/A".

If an entry is marked with one asterisk, the resource type is supported by Patterns only by utilizing the **Google Asset inventory** pattern's inclusion list. For more information, see .

If an entry is marked with two asterisks, the resource type is supported by Service Graph Connectors only by utilizing the [sn_cmdb_int_util_service_graph_resource_inclusion_whitelist] table.

Discovery and Service Mapping Patterns vs Service Graph Connector for GCP

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
admissionregistration.k8s.io/mutatingwebhookconfiguration	Yes*	Yes**	cmdb_ci_cmp_resource
admissionregistration.k8s.io/validatingwebhookconfiguration	Yes*	Yes**	cmdb_ci_cmp_resource
alloydb.googleapis.com/Cluster	Yes	No	cmdb_ci_cloud_db_cluster
apigateway.googleapis.com/api	Yes*	Yes**	cmdb_ci_cmp_resource
apigateway.googleapis.com/apiconfig	Yes*	Yes**	cmdb_ci_cmp_resource
apigee.googleapis.com/instance	Yes*	Yes**	cmdb_ci_cmp_resource
apigee.googleapis.com/organization	Yes*	Yes**	cmdb_ci_cmp_resource
apikeys.googleapis.com/key	Yes*	Yes**	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
appengine.googleapis.com/application	Yes*	Yes**	cmdb_ci_cmp_resource
apps.k8s.io/daemonset	Yes*	Yes**	cmdb_ci_cmp_resource
apps.k8s.io/Deployment	Yes*	Yes	cmdb_ci_kubernetes_deployment†
apps.k8s.io/ReplicaSet	Yes*	Yes	cmdb_ci_kubernetes_replicaset
apps.k8s.io/statefulset	Yes*	Yes**	cmdb_ci_cmp_resource
artifactregistry.googleapis.com/dockerimage	Yes*	Yes**	cmdb_ci_cmp_resource
artifactregistry.googleapis.com/repository	Yes*	Yes**	cmdb_ci_cmp_resource
bigquery.googleapis.com/bigquery-public-data/datasets	Yes	No	cmdb_ci_cloud_public_database
bigquery.googleapis.com/bigquery-public-data/datasets/tables	Yes	No	cmdb_ci_database

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
bigquery.googleapis.com/Dataset	Yes	N/A	cmdb_ci_cloud_database
bigquery.googleapis.com/Dataset	N/A	Yes**	cmdb_ci_cmp_resource
bigquery.googleapis.com/Table	Yes	No	cmdb_ci_database
bigtableadmin.googleapis.com/Instance	Yes	Yes	cmdb_ci_cloud_database
bigtableadmin.googleapis.com/Table	Yes	No	cmdb_ci_database
cloudasset.googleapis.com/feed	Yes*	Yes**	cmdb_ci_cmp_resource
cloudbilling.googleapis.com/billingaccount	Yes*	Yes**	cmdb_ci_cmp_resource
cloudbilling.googleapis.com/projectbillinginfo	Yes*	Yes**	cmdb_ci_cmp_resource
cloudfunctions.googleapis.com/CloudFunction	Yes*	Yes	cmdb_ci_cloud_function

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
cloudresourcemanager.googleapis.com/Folder	Yes	Yes	cmdb_ci_gcp_folder
cloudresourcemanager.googleapis.com/Organization	Yes	Yes	cmdb_ci_cloud_org
cloudresourcemanager.googleapis.com/Project	Yes	Yes	cmdb_ci_gcp_project
cloudresourcemanager.googleapis.com/tagkey	Yes*	Yes	cmdb_ci_cmp_resource
cloudresourcemanager.googleapis.com/tagvalue	Yes*	Yes**	cmdb_ci_cmp_resource
compute.googleapis.com/Address	Yes	No	cmdb_ci_cloud_public_ipaddresses
compute.googleapis.com/autoscaler	Yes*	Yes**	cmdb_ci_cmp_resource
compute.googleapis.com/BackendService	N/A	Yes	cmdb_ci_cloud_load_balancer
compute.googleapis.com/BackendService	Yes	N/A	cmdb_ci_lb_service

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
compute.googleapis.com/Disk	Yes	Yes	cmdb_ci_storage_volume
compute.googleapis.com/DiskTypes	Yes	No	cmdb_ci_disk_type
compute.googleapis.com/Firewall	N/A	Yes	cmdb_ci_compute_security_group
compute.googleapis.com/Firewall	Yes	Yes	cmdb_ci_network_acl_rule
compute.googleapis.com/ForwardingRule	Yes	Yes	cmdb_ci_lb_service
compute.googleapis.com/HttpHealthCheck	Yes	Yes	cmdb_ci_lb_health_service
compute.googleapis.com/Image	Yes	No	cmdb_ci_os_template
compute.googleapis.com/Instance	Yes	N/A	cmdb_ci_lb_pool_member
compute.googleapis.com/Instance	N/A	Yes	cmdb_ci_server

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
compute.googleapis.com/Instance	Yes	Yes	cmdb_ci_vm_instance
compute.googleapis.com/Instance/networkInterfaces	Yes	No	cmdb_ci_endpoint_vnic
compute.googleapis.com/Instance/networkInterfaces	Yes	Yes	cmdb_ci_nic
compute.googleapis.com/InstanceGroup	Yes	N/A	cmdb_ci_lb_pool
compute.googleapis.com/InstanceGroup	N/A	Yes**	cmdb_ci_compute_resource
compute.googleapis.com/Instancegroupmanager	Yes*	Yes**	cmdb_ci_compute_resource
compute.googleapis.com/Instancesettings	Yes*	Yes**	cmdb_ci_compute_resource
compute.googleapis.com/InstanceTemplate	Yes	No	cmdb_ci_instance_template

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
compute.googleapis.com/Instancetemplate	No	Yes**	cmdb_ci_cmp_resource
compute.googleapis.com/MachineImage	No	Yes	cmdb_ci_os_template
compute.googleapis.com/MachineTypes	Yes	No	cmdb_ci_cloud_hardware_type
compute.googleapis.com/MachineTypes	Yes	No	cmdb_ci_compute_template
compute.googleapis.com/Network	Yes	Yes	cmdb_ci_network
compute.googleapis.com/Networkendpointgroup	Yes*	Yes**	cmdb_ci_cmp_resource
compute.googleapis.com/Regions	Yes	No	cmdb_ci_google_datacenter
compute.googleapis.com/resourcepolicy	Yes*	Yes**	cmdb_ci_cmp_resource
compute.googleapis.com/router	Yes*	Yes**	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
compute.googleapis.com/Snapshot	Yes	Yes	cmdb_ci_storage_vol_snapshot
compute.googleapis.com/ssh	Yes	No	cmdb_ci_cloud_key_pair
compute.googleapis.com/sslcertificate	Yes*	Yes**	cmdb_ci_cmp_resource
compute.googleapis.com/Subnetwork	Yes	Yes	cmdb_ci_cloud_subnet
compute.googleapis.com/targetHttpProxy	Yes	N/A	cmdb_ci_lb_service
compute.googleapis.com/targetHttpProxy	Yes	Yes**	cmdb_ci_cmp_resource
compute.googleapis.com/TargetPool	Yes	N/A	cmdb_ci_cloud_load_balancer
compute.googleapis.com/TargetPool	N/A	Yes	cmdb_ci_lb_pool
compute.googleapis.com/targetSslProxy	Yes	No	cmdb_ci_lb_service

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
compute.googleapis.com/targetTcpProxy	Yes	No	cmdb_ci_lb_service
compute.googleapis.com/UrlMap	Yes	N/A	cmdb_ci_cloud_load_balancer
compute.googleapis.com/UrlMap	N/A	Yes**	cmdb_ci_cmp_resource
compute.googleapis.com/zones	Yes	No	cmdb_ci_availability_zone
container.googleapis.com/Cluster	Yes	Yes	cmdb_ci_kubernetes_cluster
container.googleapis.com/NodePool	No	Yes	cmdb_ci_kubernetes_node_pool
containerregistry.googleapis.com/image	Yes*	Yes**	cmdb_ci_cmp_resource
dataplex.googleapis.com/entrygroup	Yes*	Yes**	cmdb_ci_cmp_resource
firebase.googleapis.com/firebaseproject	Yes*	Yes**	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
firebasedatabase.googleapis.com/instances	Yes	No	cmdb_ci_cloud_database
firebaserules.googleapis.com/release	Yes*	Yes**	cmdb_ci_cmp_resource
firebaserules.googleapis.com/ruleset	Yes*	Yes**	cmdb_ci_cmp_resource
firestore.googleapis.com/Database	Yes	Yes	cmdb_ci_cloud_database
iam.googleapis.com/role	Yes*	Yes**	cmdb_ci_cmp_resource
iam.googleapis.com/serviceaccount	Yes*	Yes**	cmdb_ci_cmp_resource
iam.googleapis.com/serviceaccountkey	Yes*	Yes**	cmdb_ci_cmp_resource
integrations.googleapis.com/integration	Yes*	Yes**	cmdb_ci_cmp_resource
integrations.googleapis.com/integrationversion	Yes*	Yes**	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
K8s.io/endpoints	Yes*	Yes**	cmdb_ci_cmp_resource
K8s.io/ Namespace	Yes*	Yes	cmdb_ci_kubernetes_namespace
k8s.io/Node	Yes*	Yes	cmdb_ci_kubernetes_node
K8s.io/persistentvolume	Yes*	Yes**	cmdb_ci_cmp_resource
k8s.io/persistentvolume claim	Yes*	Yes**	cmdb_ci_cmp_resource
K8s.io/Pod	Yes*	Yes	cmdb_ci_docker_container
K8s.io/Pod	N/A	Yes	cmdb_ci_docker_image
K8s.io/Pod	N/A	Yes	cmdb_ci_kubernetes_pod
K8s.io/Pod	N/A	Yes	cmdb_ci_kubernetes_volume
K8s.io/resourcequota	Yes*	Yes**	cmdb_ci_cmp_resource
k8s.io/Service	Yes*	Yes	cmdb_ci_kubernetes_service
K8s.io/serviceaccount	Yes*	Yes**	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
labels	Yes	No	cmdb_key_value
logging.googleapis.com/logbucket	Yes*	Yes**	cmdb_ci_cmp_resource
logging.googleapis.com/logmetric	Yes*	Yes**	cmdb_ci_cmp_resource
logging.googleapis.com/logsink	Yes*	Yes**	cmdb_ci_cmp_resource
logging.googleapis.com/recentquery	Yes*	Yes**	cmdb_ci_cmp_resource
memcache.googleapis.com/Instance	Yes*	Yes	cmdb_ci_cloud_database
monitoring.googleapis.com/alertpolicy	Yes*	Yes**	cmdb_ci_cmp_resource
monitoring.googleapis.com/dashboard	Yes*	Yes**	cmdb_ci_cmp_resource
monitoring.googleapis.com/notificationchannel	Yes*	Yes**	cmdb_ci_cmp_resource
networkmanagement.googleapis.com	Yes*	Yes**	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
s.com/connectivitytest			
orgpolicy.googleapis.com/policy	Yes*	Yes**	cmdb_ci_cmp_resource
policy.k8s.io/poddisruptionbudget	Yes*	Yes**	cmdb_ci_cmp_resource
pubsub.googleapis.com/subscription	Yes*	Yes**	cmdb_ci_cmp_resource
pubsub.googleapis.com/topic	Yes*	Yes**	cmdb_ci_cmp_resource
rbac.authorization.k8s.io/ClusterRole	No	Yes	cmdb_ci_kubernetes_cluster_role
rbac.authorization.k8s.io/ClusterRoleBinding	No	Yes	cmdb_ci_kubernetes_cluster_role_binding
rbac.authorization.k8s.io/role	Yes*	Yes**	cmdb_ci_cmp_resource
rbac.authorization.k8s.io/rolebinding	Yes*	Yes**	cmdb_ci_cmp_resource
redis.googleapis.com/clusters	Yes	No	cmdb_ci_cloud_db_cluster

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
redis.googleapis.com/Instance	No	Yes	cmdb_ci_cloud_database
run.googleapis.com/revision	Yes*	Yes**	cmdb_ci_cmp_resource
run.googleapis.com/service	Yes*	Yes**	cmdb_ci_cmp_resource
servicedirectory.googleapis.com/endpoint	Yes*	Yes**	cmdb_ci_cmp_resource
servicedirectory.googleapis.com/namespace	Yes*	Yes**	cmdb_ci_cmp_resource
servicedirectory.googleapis.com/service	Yes*	Yes**	cmdb_ci_cmp_resource
servicemanagement.googleapis.com/managedservice	Yes*	Yes**	cmdb_ci_cmp_resource
servicenetworking.googleapis.com/connection	Yes*	Yes**	cmdb_ci_cmp_resource
serviceusage.googleapis.com/service	Yes*	Yes**	cmdb_ci_cmp_resource

Resource type	Supported by Cloud Discovery and Service Mapping Patterns	Supported by Cloud Service Graph Connectors	CI Class
spanner.googleapis.com/Database	Yes	No	cmdb_ci_database
spanner.googleapis.com/Instance	Yes	Yes	cmdb_ci_cloud_database
sqladmin.googleapis.com/backup	Yes*	Yes**	cmdb_ci_cmp_resource
sqladmin.googleapis.com/backuprun	Yes*	Yes**	cmdb_ci_cmp_resource
sqladmin.googleapis.com/Instance	Yes*	Yes	cmdb_ci_cloud_database
sqladmin.googleapis.com/Instances/databases	Yes*	No	cmdb_ci_database
storage.googleapis.com/Bucket	N/A	Yes	cmdb_ci_cloud_object_storage
storage.googleapis.com/Bucket	Yes	N/A	cmdb_ci_cmp_resource
storage.k8s.io/storageclass	Yes*	Yes**	cmdb_ci_cmp_resource

Accessing the connection details of Service Graph Connector for GCP

You can access the connection details of the Service Graph Connector for GCP in a single view using the common connection framework (CCF) included within the Integration Commons for CMDB (sn_cmdb_int_util) store app.

With the CCF, you can access all the connections used by the Service Graph Connector for GCP. The connection details include the connection alias, connection properties, data sources, and scheduled data imports associated with a connection. You can also test the connection. For more information, see [Accessing the connection details of Service Graph Connectors](#).

Access the details of a GCP connection

Access the details of a GCP connection configured for the Service Graph Connector for GCP.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Service Graph Connectors > GCP > Connections**.
2. From the **Name** column of the Service Graph Connections list, select a connection.
3. On the Service Graph Connections page, view the connection details such as the connection name, alias, and data sources.
4. Select a related list to view further details of the connection.
5. (Optional) Select the **Test Connection** related link to test the connection.

Handling import failures for large GCP resources

Manage the data source failure when importing resources from GCP.

When importing large GCP resources, the data sources in the Service Graph Connector for GCP might fail because of a Direct REST response exceeding the 1024-MB limit. For more information, see [REST step](#).

You can use one or more of the following options to handle the REST response size limitation error in the Service Graph Connector for GCP:

- [Save the response as an attachment](#)
- [Modify the batch size](#)

Save the response as an attachment

With the save response as an attachment feature, the response of each data source is temporarily stored as an attachment and after the response is read, the attachment is deleted automatically. The save response as an attachment feature is activated by default and is triggered by the `sn_gcp_integ.save_rest_response_as_attachment` system property.

As a user with the admin role, you can deactivate the save response as an attachment feature by setting the `sn_gcp_integ.save_rest_response_as_attachment` property to `false`. When the feature is deactivated and the response size is within the limit, the data sources run normally.

Modify the batch size

By default, the maximum batch size of the environment variable for a data source is set to 100 records. You can reduce or increase the batch size based on the data volume. To manage the batch size of a data source, you must add a system property of integer type for the data source as described in the following table.

System properties for handling the batch size of Service Graph Connector for GCP data sources

Data source	System property
SG-GCP Cloud Database	<code>sg_gcp_cloud_db_batch_size</code>
SG-GCP Cloud Function	<code>sg_gcp_cloud_function_batch_size</code>

Data source	System property
SG-GCP Cloud Object Storage	sg_gcp_object_storage_batch_size
SG-GCP Kubernetes Cluster	sg_gcp_kubernetes_cluster_batch_size
SG-GCP Kubernetes Cluster Role	sn_gcp_integ.sg_gcp_kubernetes_cluster_role_batch_size
SG-GCP Kubernetes Cluster Role Binding	sn_gcp_integ.sg_gcp_kubernetes_cluster_role_binding_batch_size
SG-GCP Kubernetes Deployment	sg_gcp_kubernetes_deployment_batch_size
SG-GCP Kubernetes Namespace	sg_gcp_kubernetes_namespace_batch_size
SG-GCP Kubernetes Node	sg_gcp_kubernetes_node_batch_size
SG-GCP Kubernetes Pod	sg_gcp_kubernetes_pod_batch_size
SG-GCP Kubernetes Replicaset	sg_gcp_kubernetes_replicaset_batch_size
SG-GCP Kubernetes Service	sg_gcp_kubernetes_service_batch_size
SG-GCP Load Balancer	sg_gcp_load_balancer_batch_size
SG-GCP Load Balancer Health Service	sg_gcp_lb_health_service_batch_size
SG-GCP Load Balancer Pool	sg_gcp_lb_pool_batch_size
SG-GCP Load Balancer Pool Member	sg_gcp_lb_pool_member_batch_size

Data source	System property
SG-GCP Load Balancer Service	sn_gcp_integ.sg_gcp_lb_service_batch_size
SG-GCP Network	sg_gcp_network_batch_size
SG-GCP Kubernetes Node Pool	sn_gcp_integ.sg_gcp_kubernetes_node_pool_batch_size
SG-GCP Security Group	sg_gcp_security_group_batch_size
SG-GCP Software Inventory	sg_gcp_software_inventory_batch_size
SG-GCP Storage Volume	sg_gcp_storage_volume_batch_size
SG-GCP Storage Volume Snapshot	sg_gcp_storage_vol_snapshot_batch_size
SG-GCP Subnet	sg_gcp_subnet_batch_size
SG-GCP VM Instance	sg_gcp_vm_instance_batch_size

Note: You must add these system properties as they aren't available by default. You must have the admin role to add or modify a system property. For more information, see [Add a system property](#).

Deep discovery for Service Graph Connector for GCP

The SG-GCP Execute Patch Job and SG-GCP Get Patch Job scheduled import jobs enable deep discovery by running patch jobs at the GCP project level.

Deep discovery involves fetching server data. The deep discovery data populates host name, serial number, CPU data, TCP connections, and running processes data into the CMDB.

The Service Graph Connector for GCP uses OS Config Patch jobs on GCP projects to execute the deep discovery commands on the VMs. These patch jobs run independently on Linux and Windows VMs.

The following scheduled imports execute the OS Config Patch jobs, and then read and delete the output:

- [SG-GCP Execute Patch Job](#)
- [SG-GCP Get Patch Job](#)

Deep discovery connection properties

Deep discovery connection properties

Connection property name	Connection property value
Scripts Bucket Name	Name of the Cloud Storage Bucket where the deep discovery scripts are saved.
Outputs Bucket Name	Name of the output Cloud Storage Bucket.
Outputs Storage Path	Path of the output Cloud Storage Bucket.
Shell Script Object Path	Path of the deep discovery Shell script file.
Shell Script Generation Number	Number that is used to identify the versioning of the cloud storage object.
PowerShell Script Object Path	Path of the deep discovery PowerShell script file.
PowerShell Script Generation Number	Number that is used to identify the versioning of the cloud storage object.

SG-GCP Execute Patch Job

The SG-GCP Execute Patch Job scheduled import reads the path of the Shell and PowerShell scripts uploaded to the Cloud Storage Bucket and the generation numbers from the deep discovery connection properties.

The scheduled import then runs the commands on Linux and Windows VMs separately.

In the Import Set table for the SG-GCP Execute Patch Job import schedule, each project will have two rows, one for Linux VMs and the other for Windows VMs. The **Import Set** rows remain in Pending state because no transformations are required for these rows. The **Import Set** rows include the patch job ID and the list of instances on which the deep discovery script is run. The output is stored in the Cloud Storage Bucket with the Object ID of the VMs on which the deep discovery commands are run.

SG-GCP Get Patch Job

The SG-GCP Get Patch Job scheduled import reads the output of the commands that are run on the Linux and Windows VMs. The scheduled import reads the output from the output bucket and folder path that are specified in the deep discovery connection properties. The SG-GCP Get Patch Job scheduled import then deletes the output from the Cloud Storage Bucket. The **Import Set** rows are processed using Application Dependency Mapping (ADM) to populate the Server records with the data of the VMs on which deep discovery was run. The **Import Set** rows include VM data such as running processes, TCP connections, and CPU information.

Application Dependency Mapping

Using the deep discovery information, Application Dependency Mapping (ADM) creates and classifies the following applications from the imported middleware information:

- Apache Web Server
- MongoDB Instance
- MSFT SQL Instance
- MySQL Instance
- Nginx Web Server
- Oracle Database Listener
- Oracle Instance

- PostgreSQL Instance
- Tomcat

Additional resource

[Service Graph Connector for GCP - Deep Discovery Setup Instructions \[KB2213095\]](#) article in the Now Support Knowledge Base

Life cycle management of records in Service Graph Connector for GCP

Life cycle management in the in the Service Graph Connector for GCP monitors and updates the statuses of GCP resources throughout their entire life cycle, from creation to deletion.

The life cycle management process helps maintain the accuracy and integrity of data in the Configuration Management Database (CMDB).

In life cycle management, the record removal process involves systematically deleting obsolete or unnecessary resources. This step ensures that outdated entries are cleared, keeping the CMDB accurate and up-to-date. See [Record removal process in Service Graph Connector for GCP](#).

Life cycle management for CIs in Service Graph Connector for GCP

The following table lists the configuration items (CIs) for which life cycle management is available in Service Graph Connector for GCP.

Supported CIs for life cycle management

Data source	CMDB CI classes	Life cycle management available
SG-GCP Organization	Cloud Organizations [cmdb_ci_cloud_org]	Yes
SG-GCP Folder	Google Organization Folder [cmdb_ci_gcp_folder]	Yes

Data source	CMDB CI classes	Life cycle management available
SG-GCP Project	Google Organization Project [cmdb_ci_gcp_project]	Yes
	Cloud Service Account [cmdb_ci_cloud_service_account]	Yes
SG-GCP Network	Cloud Network [cmdb_ci_network]	Yes
SG-GCP Machine Image	Image [cmdb_ci_os_template]	Not applicable
SG-GCP Subnet	Cloud Subnet [cmdb_ci_cloud_subnet]	Yes
	Google Datacenter [cmdb_ci_google_datacenter]	Not applicable
SG-GCP Storage Volume	Storage Volume [cmdb_ci_storage_volume]	Yes
SG-GCP Storage Volume Snapshot	Storage Volume Snapshot [cmdb_ci_storage_volume_snapshot]	Yes
SG-GCP Security Group	Compute Security Group [cmdb_ci_compute_security_group]	Yes

Data source	CMDB CI classes	Life cycle management available
	Network ACL [cmdb_ci_network_acl]	Yes
	Network ACL Rule [cmdb_ci_network_acl_rule]	Yes
SG-GCP Software Inventory	Software [cmdb_ci_spkg] Software Instance [cmdb_software_instance] Software Installation [cmdb_sam_sw_install]	Yes
SG-GCP VM Instance	Virtual Machine Instance [cmdb_ci_vm_instance]	Yes
	Server [cmdb_ci_server]	Yes
	Storage Mapping [cmdb_ci_storage_mapping]	Yes
	VNIC Endpoint [cmdb_ci_endpoint_vnic]	Yes
	Block Endpoint [cmdb_ci_endpoint_block]	Yes

Data source	CMDB CI classes	Life cycle management available
	IP Address [cmdb_ci_ip_address]	Yes
	Cloud Mgmt Network Interface [cmdb_ci_nic]	Yes
	Hardware Type [cmdb_ci_compute_template]	Not applicable
	Image [cmdb_ci_os_template]	Yes
	Availability Zone [cmdb_ci_availability_zone]	Not applicable
SG-GCP Execute Patch Job	Not applicable	Not applicable
SG-GCP Hardware Type	Hardware Type [cmdb_ci_compute_template]	Not applicable
	Google Datacenter [cmdb_ci_google_datacenter]	Not applicable
	Availability Zone [cmdb_ci_availability_zone]	Not applicable
SG-GCP VM Hw Consolidation	Virtual Machine Instance [cmdb_ci_vm_instance]	Not applicable

Data source	CMDB CI classes	Life cycle management available
SG-GCP Load Balancer Pool	Load Balancer Pool [cmdb_ci_lb_pool]	Yes
SG-GCP load Balancer Pool Member	Load Balancer Pool Member [cmdb_ci_lb_pool_member]	Yes
SG-GCP Load Balancer Health Service	Cloud Load Balancer Health Service [cmdb_ci_lb_health_service]	Yes
SG-GCP Load Balancer	Cloud Load Balancer [cmdb_ci_cloud_load_balancer]	Yes
SG-GCP Load Balancer Service	Load Balancer Service [cmdb_ci_lb_service]	Yes
SG-GCP Cloud Database	Cloud DataBase [cmdb_ci_cloud_database]	Yes
SG-GCP Cloud Function	Cloud Function [cmdb_ci_cloud_function]	Yes
SG-GCP Cloud Object Storage	Cloud Object Storage [cmdb_ci_cloud_object_storage]	Yes
SG-GCP Kubernetes Cluster	Kubernetes Cluster [cmdb_ci_kubernetes_cluster]	Yes
SG-GCP Kubernetes Node	Kubernetes Node [cmdb_ci_kubernetes_node]	Yes

Data source	CMDB CI classes	Life cycle management available
SG-GCP Kubernetes Pod	Kubernetes Pod [cmdb_ci_kubernetes_pod]	Yes
	Kubernetes Volume [cmdb_ci_kubernetes_volume]	No
	Docker Image [cmdb_ci_docker_image]	No
	Docker Container [cmdb_ci_docker_container]	Yes
SG-GCP Kubernetes Service	Kubernetes Service [cmdb_ci_kubernetes_service]	Yes
SG-GCP Kubernetes Namespace	Kubernetes Namespace [cmdb_ci_kubernetes_namespace]	Yes
SG-GCP Kubernetes Deployment	Kubernetes Deployment [cmdb_ci_kubernetes_deployment]	Yes
SG-GCP Kubernetes Replicaset	Kubernetes ReplicaSet [cmdb_ci_kubernetes_replicaset]	Yes
SG-GCP Kubernetes Cluster Roles	Kubernetes Cluster Role [cmdb_ci_kubernetes_cluster_role]	Yes

Data source	CMDB CI classes	Life cycle management available
SG-GCP Kubernetes Cluster Role Binding	Kubernetes Cluster Role Binding [cmdb_ci_kubernetes_cluster_role_binding]	Yes
SG-GCP Kubernetes Node Pool	Kubernetes Node Pool [cmdb_ci_kubernetes_node_pool]	Yes
SG-GCP Generic Resource	Cloud Resource [cmdb_ci_cmp_resource]	Yes
SG-GCP Annotation	Key Value [cmdb_key_value]	Not applicable
SG-GCP Get Patch Job	Not applicable	Not applicable
SG-GCP Generic Resource Annotation	Key Value [cmdb_key_value]	Not applicable

Record removal process in Service Graph Connector for GCP

The Service Graph Connector for GCP uses soft deletion and hard deletion to manage the removal of records.

Record removal is the process of handling data that is no longer needed. For any discovered resources that are deleted later after pulling data, the Service Graph Connector for GCP automatically updates the **Install Status** field of the associated CMDB CI classes to indicate any retired or deleted records.

- [Soft deletion](#)
- [Hard deletion](#)

Soft deletion

Soft deletion involves marking the install status of a record as **retired** rather than permanently deleting it from the system. The record remains in the database, making it possible to reference or restore it later.

Starting with the Service Graph Connector for GCP 1.11.0 version, Integration Commons Record Removal is implemented to mark the Install Status and Operational Status of configuration items (CIs) as retired if they aren't discovered during the latest pull. This implementation verifies the Last Scan attribute of the CIs in the Source [sys_object_source] table to identify CIs that aren't discovered during the latest pull. Additionally, the source native keys (SNKs) for all CIs are modified to include the connection alias ID to identify the connection that discovered specific CIs.

Soft deletion for the Service Graph Connector for GCP 1.11.0 version relies on the Source [sys_object_source] table and the Use Record Removal and lookback_time_in_days connection properties. Soft deletion for the Service Graph Connector for GCP 1.10 version and earlier relied on the SG-GCP Extension Attributes [sn_gcp_integ_extension_attributes] table.

After upgrading to version 1.11.0, the value of the Use Record Removal connection property is set to `true` when the import is run for the first time. Consequently, the latest SNKs for the CIs are included in the Source [sys_object_source] table. Record removal is implemented from the subsequent run.

After upgrading to version 1.11.0, the value of the lookback_time_in_days connection property is set to null by default. During the initial run after the upgrade, the Source [sys_object_source] table is queried for all records that the connection discovered. By default, the value is set to 15 after the initial run. Two weeks after the initial run, the Source [sys_object_source] table is queried for all records that weren't discovered in the past 15 days. All records that weren't discovered in the past 15 days are marked as retired.

Soft deletion is enabled for the following data sources in Service Graph Connector for GCP:

- SG-GCP Folder
- SG-GCP Project

- SG-GCP Network
- SG-GCP Subnet
- SG-GCP Storage Volume
- SG-GCP Storage Volume Snapshot
- SG-GCP Security Group
- SG-GCP VM Instance
- SG-GCP Load Balancer Pool
- SG-GCP Load Balancer Pool Member
- SG-GCP Load Balancer Health Service
- SG-GCP Load Balancer
- SG-GCP Load Balancer Service
- SG-GCP Cloud Database
- SG-GCP Cloud Function
- SG-GCP Cloud Object Storage
- SG-GCP Kubernetes Cluster
- SG-GCP Kubernetes Node
- SG-GCP Kubernetes Pod
- SG-GCP Kubernetes Service
- SG-GCP Kubernetes Namespace
- SG-GCP Kubernetes Deployment
- SG-GCP Kubernetes Replicaset
- SG-GCP Kubernetes Cluster Roles
- SG-GCP Kubernetes Cluster Role Binding
- SG-GCP Kubernetes Node Pool

Starting with the Service Graph Connector for GCP 1.10 version, the SG-GCP Generic Resource data source uses the post-script method for soft deletion, which relies on the Source [sys_object_source] table. For more information about the post-script method, see [Post-script method](#).

Hard deletion

Hard deletion involves permanently deleting a record from the system. The record is removed from the CMDB and can't be referenced or restored later. The SG-GCP Software Inventory data source uses the hard deletion method for the deletion of records.